

CLM Process Operations Manual

LK11-340

**Communication
Link Module**

***CLM Process
Operations Manual***

**LK11-340
01/09**

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About This Publication

This document is the operators manual for the TotalPlant Network (TPN) Communication Link Module (CLM).

The CLM provides a standard communications platform which allows the TPN to interface with a wide variety of devices/networks. With this interface, the operator has access to points resident on non-Honeywell devices that are connected to non-TPN networks. Access to these devices and networks is transparent so non-Honeywell devices appear to be directly connected to the Local Control Network (LCN).

There are two versions available for the CLM . Both are completely supported by this operations manual.

- **CLM** is the original CLM which is installed to an AM node and requires the CLM communications boards, CLMI or CCP.

Where there are references to hardware within this manual, such as: hardware, slots, boards, ports, device, etc., in the CLM environment these items are **physical**.

Note: References to CLMI (CLMI interface board) are generally inclusive of the CLMI, CCP, and UCIM interface boards except as indicated above.

- **TPS/CLM** installs to the APP node with an AM personality installed and emulates the previous CLMI/CCP communications boards used in the earlier version of the CLM. The concept of these boards and communications ports is carried over and emulated in the TPS/CLM, however, the communications capability is now supported in the Windows NT environment and may include either physical or logical (emulated) communications devices.

Where there are references to hardware within this manual, such as: hardware, slots, CLMI/CCP boards, ports, device, etc., in the TPS/CLM environment these items may be **physical** or **emulated**.

Specifically for TPS/CLM;

- slots and boards are always **emulated**
- ports and devices may be **physical or emulated** depending on the application

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Section 1 – Communication Link Module Operations

1.1 CLM Overview

Overview

This section provides a detailed description of the available CLM operating displays used in CLM operations.

TPN operator interface

The CLM provides the TPN operator a window of access to points located on a non-TPN network. The CLM interfaces with these points in a manner that simulates their direct connection to the TPN Local Control Network (LCN).

CLM environment

The CLM provides the mechanism for the TPN to communicate with a wide variety of non-TPN networks. Two CLM environments are supported:

- **CLM** is the original CLM which is installed to an AM node and requires the CLM communications boards CLMI, CCP, or UCIM.
- **TPS/CLM** is installed on an APP node with an AM personality installed and ‘emulates’ the CLMI/CCP hardware communications boards and slots used in the earlier version of the CLM. In this environment, ports and devices may be ‘emulated’ or physical and are application specific. The user operation of the interface is the same as the original CLM.

The CLM modifies a standard Applications Module (AM) personality, with additional software. Additional software (and/or hardware) facilitates connection of the CLM to a non-Honeywell device or network (hereafter referred to as the non-Honeywell device or non-TPN network), resulting in a flexible communications interface.

The interface uses AM points with Custom Data Segments (CDS) to define image points in the CLM that represent points in the non-Honeywell network's database. Specific to the CLM application installed, the operator can monitor and change selected operating parameters in the remote point by changing the associated parameters in the CLM image point.

1.2 Introduction to the Displays

TPS/CLM Displays

All operating displays described here apply to both the TPS/CLM and standard CLM. The TPS/CLM has additional displays used for installation, configuration, and status within the Windows NT environment.

Configuring multiple CLMs

Multiple CLMs can be configured within a TPN System. Each CLM is identified by its entity ID, CLM_*nn* where *nn* is the TPN CLM Node number (i.e., CLM_55).

Each CLM contains its own Network Memory that includes all of the CLM interface structures required to support the operating realm of that CLM. Therefore, if multiple CLMs are configured within your TPN system each CLM represents an independent node that can communicate with other TPN nodes using the standard capabilities of the TPN.

The CLM provides the operator access to points located on a non-TPN network in a manner that makes these points appear to be directly connected to the TPN. This implementation allows the standard LCN Alarm, Group, Detail, and Custom display capabilities to be used to display data acquired by the CLM.

A display package is provided with the CLM Platform for monitoring and control of the non-TPN network and its devices.

CLM display structure

The top-down structure in Figure 1-1 shows the displays that are available with the CLM Platform.

After the initial display call up of CLMSCHEM, all other displays are accessed by selecting a target or pressing a dedicated key (i.e. PAGE FWD, PAGE BACK) on the current display.

This top-down display structure allows you to move logically from display to display, obtaining more detailed information and performing functions as needed to operate your application. A detailed description of each of these displays is included in the sections that follow.

WARNING

The "View" mode of the User Station does not allow updates of any kind. Therefore, the "View" mode is not supported.

The unit to which the CLM_*nn* and BUILD_*nn* points are assigned, during CLM installation, must be configured on the console which utilizes the CLM displays.

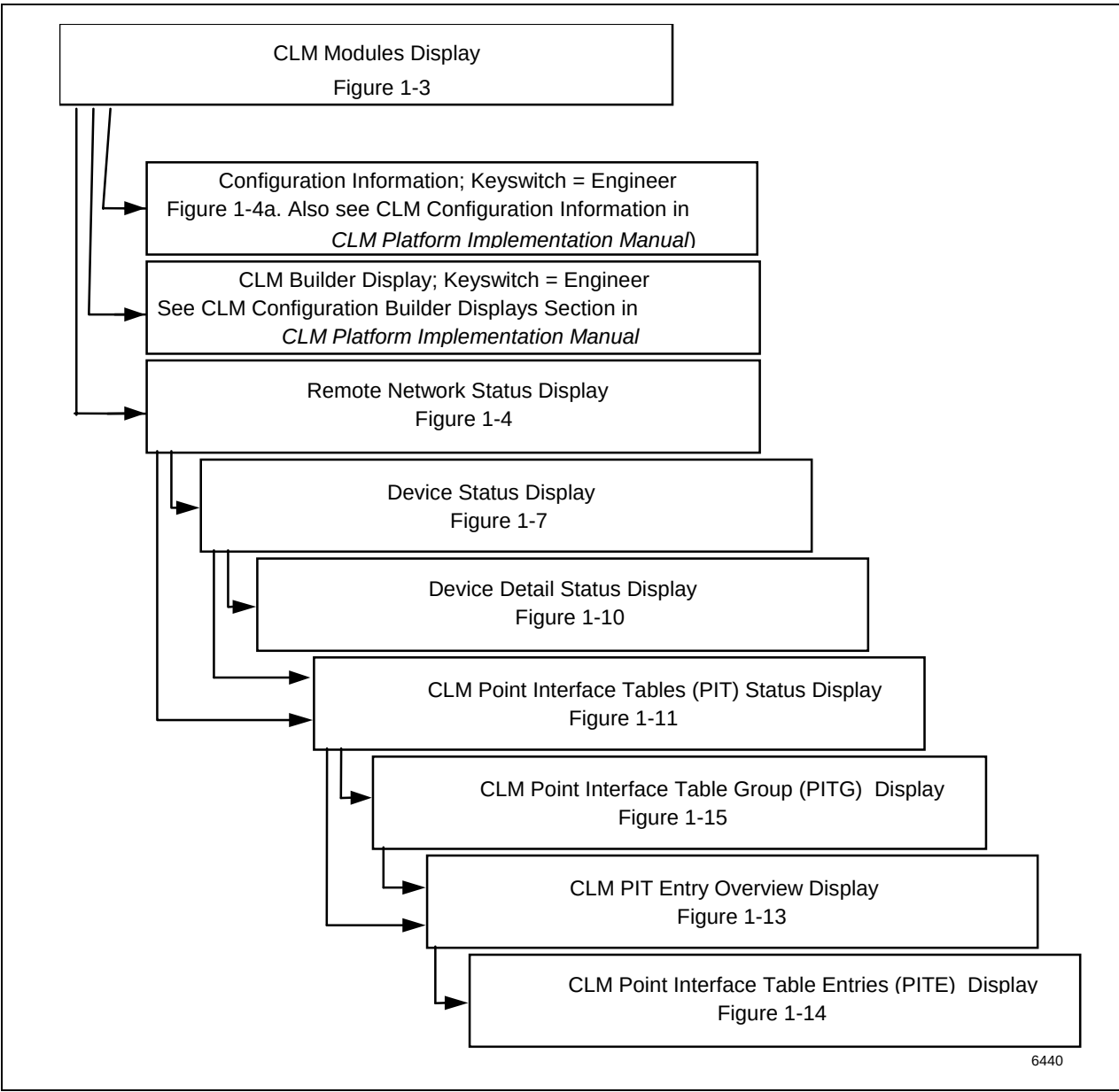
When the engineer key is used it allows the override of either of these conditions. Therefore, the engineer should exit from all CLM displays prior to the removal of the key.

Cross screen action is not supported in the CLM displays.

Continued on next page

1.2 Introduction to the Displays, Continued

Figure 1-1 CLM Display Structure



CLM Display Navigation Summary, Parts 1 and 2

Figures 1-2 and 1-9, CLM Display Navigation Summary - Parts 1 and 2, are structure tables that show navigation paths through the displays.

1.3 CLM Display Navigation Summary

Structure table

Figure 1-2 is a structure table that shows the possible navigation paths through the CLM displays. Each row in the table depicts a current display and the paths available to other displays. For example, the fourth row should be read as follows:

“IF the current display is Remote Network Status, Figure 1-6, AND a specific **PORT** target is selected, AND the **DETAIL STATUS** target is selected, THEN the new display presented will be Device Status, Figure 1-7.”

Figure 1-2 CLM Display Navigation Summary, Part 1

Display navigation summary, part 1

The left column contains every display from which you can access another display. For any given row, the middle two columns define the targets that must be selected in order to show the display in the right column of the same row.

Note: Part 2 of this CLM Display Navigation Summary is shown in Figure 1-9.

IF The Current Display (Figure #nn) Is . . .	AND The Item Selected Is . . .	AND The Function Target Selected Is . . .	THEN The New Display To Be Presented Will Be . . .
CLM Modules (Fig. 1-3)	CLM Node		Remote Network Status (Fig. 1-4)
Remote Network Status (Fig. 1-4)	CLMI _____ PORT	_____	Remote Network Status (Fig. 1-5) _____ Remote Network Status (Fig. 1-6)
Remote Network Status (Fig. 1-6)	PORT	DETAIL STATUS _____ PIT STATUS	Network Device Status (Fig. 1-7) _____ PIT Status (Fig. 1-11)

Navigation Targets

These navigation targets are common to all CLM displays.

- The **DISPLAY BACK** returns to the previous CLM display.
- The **PAGE BACK** applies to multi page displays, if applicable, to return to the previous page of the display.
- The **PAGE FWD** applies to multi page displays, if applicable, to go to the next page of the display.

Note 1: The **DISPLAY BACK** , **PAGE BACK** , and **PAGE FWD** functions are also available using the operator or engineering keyboard buttons.

Note 2: The **DISPLAY BACK** , **PAGE BACK** , and **PAGE FWD** targets appear only when applicable.

1.4 Bringing Up the CLM Modules Display (CLMSCHEM)

Two possibilities

The CLM Modules display is called by selecting the assigned keyboard button (if assigned on your keyboard), OR by following one of these the procedures.

Note: This is the only CLM display that is called by entering a schematic name; all other CLM displays are activated from defined targets on the CLM displays.

Accessing the CLM Modules Display – Primary access.

Step	Action
1	Press SCHEM on the operator's keyboard.
2	System will respond: Enter Schematic Name
3	Key in: CLMSCHEM then ENTER – CLMSCHEM is displayed. Fig. 1-3

Accessing the CLM Modules Display – Alternate access **R400** and earlier.

Step	Action
1	Press the System Status button on the operator's keyboard.
2	Select the Application Modules target.
3	Select the STATUS DETAIL target at the bottom of the page. -- The system responds by displaying the applicable status objects
4	Select the object line with CLM MODULE in the text. Note: A PAGE FWD may be required to display the CLM object line.
5	Select the VIEW OBJECT DETAIL target at the bottom of the page – CLMSCHEM is displayed.

1.4 Bringing Up the CLM Modules Display (CLMSCHEM)

Continued

Accessing the CLM Modules Display – Alternate access **R500** and later.

Step	Action
1	Press the System Status button on the operator's keyboard.
2	Select the CLM/AM node number on the display.
3	Select the NODE STATUS target at the bottom of the page. -- The system responds by displaying the AM Node Status display
4	Select the STATUS DETAIL target at the bottom of the page. -- The system responds by displaying the applicable status objects
5	Select the object line with CLM MODULE in the text. Note: A PAGE FWD may be required to display the CLM object line.
6	Select the VIEW OBJECT DETAIL target at the bottom of the page – CLMSCHEM is displayed.

1.5 CLM Modules Display

Using the CLM Modules Display

The CLM Modules display presents the overall status of each CLM. For each CLM Node, the specific node number, CLM description, and composite CLM communications status is displayed.

Figure 1-3 CLM Modules Display

COMMUNICATIONS LINK MODULES			SCHEM: CLMSTAT1		
NODE	DESCRIPTION	STATUS	NODE	DESCRIPTION	STATUS
52	NEW PLATFORM CLM POINT	OK			
54	NEW PLATFORM CLM POINT	OK			
*55	NEW PLATFORM CLM POINT	UNKNOWN			
56	NEW PLATFORM CLM POINT	OK			
57	CLM 57 CONTROL POINT	OK			
58	NEW PLATFORM CLM POINT	OK			
60	NEW PLATFORM CLM POINT	OK			

Operation

For additional CLM node details, select a specific CLM Node by targeting on the desired line. The Remote Network Status is displayed, Figure 1-4.

1.5 CLM Modules Display, Continued

Communications Status The Communications Status Legend, Table 1.5-1, applies to the CLM Modules display, Figure 1-3, and to the Remote Network Status display, Figures 1-4.

The same status/color legend also applies to the Device Status display, Figure 1-7.

The communication status is the second line in component targets.

Table 1.5-1 Communications Status Legend

Communication Status	Color	Description
OK	Cyan	OK: The network component is operating normally.
ERROR	Yellow	Error: A communication error was encountered, and a retry is underway. Status will change to OK if the retry is successful or FAILED if all retries are unsuccessful.
FAILED	Red	Failed: All communication attempts to the component have been unsuccessful.
ACKFAIL	Red	Acknowledged Failed: Same as FAILED except the operator has acknowledged the FAILED status. The literal ACKFAIL does not blink.
P_FAIL	Red	Partially Failed: This indicates a failure at a lower level. If at the board level then some, but not all, ports or devices are failed on that board. If at the CLM level, then one of the two boards has some lower-level failure and this CLM also shows P_FAIL. If at the device level, then a) some but not all of the subordinate devices have failed, or b) there are errors at the PITE parameter level that need to be resolved for entries associated with this device.

Continued on next page

1.5 CLM Modules Display, Continued

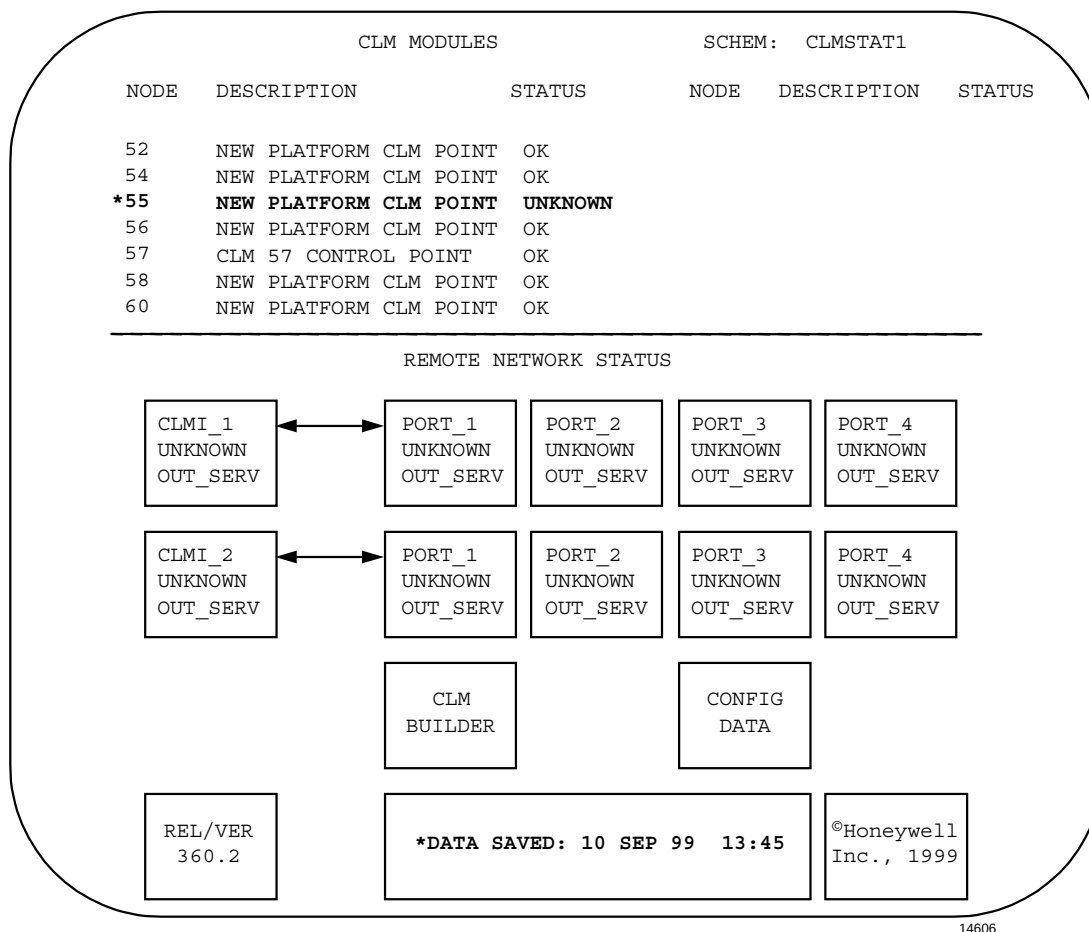
UNKNOWN	Yellow	Unknown: The current operating state of the component is out of service (OUT_SERV). See note below.
PRI_UNAV	Yellow	Primary Unavailable: A higher level network component is not available for communications, or has failed. See note below.
Note: When the communications status code is UNKNOWN or PRI_UNAV the component or its network must be placed in-service in order to continue. The in-service state change operation then changes the communications status to either OK or FAIL.		

1.6 Remote Network Status Display

Remote network status display targets

When a CLM Node number is selected on the CLM Modules display, Figure 1-3, the Remote Network Status is displayed on the lower half of the display, Figure 1-4.

Figure 1-4 Remote Network Status Display - After CLM Node selection



Note: The CLMI board and port names (top line in each target) show relative position. The actual names displayed are those assigned by your applications engineer.

- After selecting a CLM Node, you can select one of the following targets:
- A **CLMI**, this revises the display per Figure 1-5.
- A **PORT**, this revises the display per Figure 1-6.
- The **CONFIG DATA** (ENGR level) calls the CLM Configuration Information Screen, Figure 1-4a.
- The **CLM BUILDER** (ENGR level) calls the CLM Configuration Builder. Reference the *CLM Platform Implementation Manual*.

Continued on next page

1.6 Remote Network Status Display, Continued

Display layout

- **CLMI 1** target and 1 to 4 **PORT** targets on the first row
- **CLMI 2** target and 1 to 4 **PORT** targets, on the second row
- Each CLMI and PORT target displays: the network component name, communications status – Ref Table 1-3, and its current operating state.
- Targets display for each board and port component configured in the selected CLM. Any possible component not configured displays a blank target. For example, if a second board is not configured, the second row of 5 targets displays empty.

Date Saved

When the CLM node is selected, the time and date stamp of the last successful CLM checkpoint file is displayed at the bottom of the display. If there has been successful builder activity since that time, an asterisk appears next to the CLM node number and the saved data time/date stamp. This inconsistency changes the time/date stamp from cyan to yellow. If the last CLM checkpoint was not successful, the asterisk and the time/date stamp are in red.

Figure 1-4a Configuration Information CLM_nn - After CLM Node selection and CONFIG DATA

STATE CHANGE ACCESS OPERATOR SUPERVISOR ENGINEER	PATHNAME CONFIGURATION SAVE DATA NULL LOAD &CUS BUILDER SUPPORT NET > &CUS CLMI APPLICATIONS NET > &CUS
COMMUNICATIONS ERROR MESSAGE PRINT ENABLE DISABLE	
SCHEDULER CYCLE 1 Que.SIZE 4	SYNC RETRY TIME 120 SECONDS

14625

Allows an Engineer (Keylock) to review/change the configurable pathnames, device state change access level, communication error message print, scheduler values, and synchronization retry time. For more details see *CLM Platform Implementation Manual*.

Continued on next page

1.6 Remote Network Status Display, Continued

Communication Error Message Print

This function may be ENABLED or DISABLED. If disabled, journal messages that indicate a device status change from OK to ERROR, or ERROR to OK will be suppressed. They will not appear at the typer or in the History Module event retrieval. Normally, this option is used when the communication medium is highly intermittent, but no solid loss of communications occurs.

Remember! The status of a device changes to ERROR when at least 1 message is lost, but all retries have not yet been attempted.

Scheduler CYCLE

The range for the scheduler CYCLE is 0 to 10 seconds with a default of 1 second. The platform scheduler runs every schedule cycle to see if there are PITs to schedule.

Entering a value of 0 has the affect of enabling the “fast” scheduler rate. This is required if you need to configure any PITs at a 1 second scan rate. This will run the scheduler CYCLE every ½ second.

Warning! Use of the 1 second PIT through this cycle is extremely performance impactive and should be used moderately!

For a large database configuration of more than 200 PITs, a slower scheduler CYCLE is recommended to minimize CPU usage in the CLM. For example, if the PITs typically have cycles greater than 10 seconds, the scheduler could run at a cycle between 2 and 10 seconds to maximize the CLM CPU free time.

Caution: It is the engineer’s responsibility to assure that the scheduler CYCLE is always less than or equal to the PIT cycle.

Scheduler QUE_SIZE

The range for the scheduler QUE_SIZE is 4 to 20 with a default of 4. The scheduler queue size refers to the number of requests that can be queued per port. The scheduler queues PIT requests on a priority basis; i.e., if more PITs are ready to be scheduled than are designated in the QUE_SIZE configuration, the one with the highest behind count or fastest cycle time is scheduled before those that are not as far behind or have slower cycle times.

By increasing the QUE_SIZE, more requests can be queued. However, this allows more slow PITs to get in the queue ahead of faster PITs, and the faster PITs must then wait for the slower PITs. This could have the effect of blocking the faster PITs and increasing their behind counts.

For applications that have a quick PIT turnaround time, the QUE_SIZE default of 4 may not be enough. More requests can be processed within one schedule cycle if the QUE_SIZE increases. It is recommended that for schedule cycles of 1 or 2 the number for QUE_SIZE be 4 or 8 respectively. The larger the scheduler CYCLE, the more outstanding requests can be configured.

Note: Scheduler CYCLE and QUE_SIZE apply to both PITs and PITGs.

The CLM needs to be checkpointed and restarted after the changes are made for them to be effective. Remember, if redundant, both nodes must be shut down.

ATTENTION

Continued on next page

1.6 Remote Network Status Display, Continued

Device State Change - Access Level Configuration

The CHANGE STATE operation in the CLM is key access level dependent. The three states correspond to the key positions, OPR, SUP, ENGR. Selecting one of these three levels on this display will set the minimum key access level required to display the CHANGE STATE target on the Remote Network Display and Device Detail Display.

Example: If the State Change Access value is set to SUPERVISOR, a key position of OPR will suppress the display of the CHANGE STATE target, while key positions of SUP and ENG will display the target for use.

Device State Change Access Level References

Configuration	Key Position	Accessible
OPR	OPR	Yes
OPR	SUP	Yes
OPR	ENG	Yes
SUP	OPR	No
SUP	SUP	Yes
SUP	ENG	Yes
ENG	OPR	No
ENG	SUP	No
ENG	ENG	Yes

The default setting is OPERATOR. Select the level required and press the ENTER key.

Pathname Configuration

The entries for the three pathname configuration fields are used to configure data resource locations for specific CLM operations.

The SAVE DATA pathname requires a directory entry for the storage locations of the CLM Network Memory checkpoint data file. The media device will always track the AM checkpoint media selection.

The NULL LOAD target allows the user to clear the CLM database on CLM node reload.

The BUILDER SUPPORT pathname requires an entry that includes a media and directory (i.e., NET>&CUS). This pathname is the source of all Template Definition Files (TDF) used during Configuration Builder operations.

The CLMI APPLICATIONS pathname requires an entry that includes a media and directory (i.e., PN:01>RM00>VOL1). This pathname is the source of the Device Detail Status display source (.ZS), CLMI Configuration file (.ZC), CLMI and CCP operating systems (.S) and the application protocol files (.S).

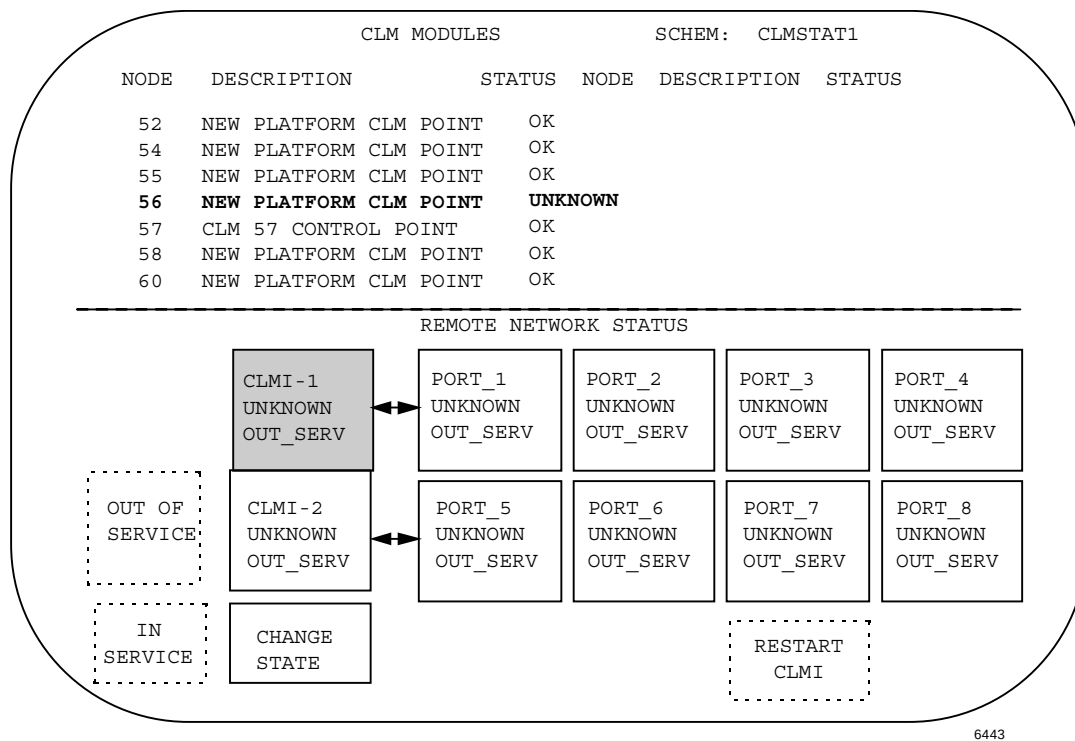
Important! All changes made to these fields, or any on the Configuration Page, MUST be followed by a SAVE DATA of the CLM to preserve the changes through node restart.

Sync Retry Time

The time to wait after a synchronization request to a component has failed before the request is issued again. The default for this value is 120 seconds.

1.6 Remote Network Status Display, Continued

Figure 1-5 Remote Network Status Display - After Specific CLMI target selection



CLMI

A specific CLMI target selection revises the display per Figure 1-5.

The **CHANGE STATE** target appears at the bottom of the display.

The current CLMI target is highlighted with a blue background upon selection. The **RESTART CLMI** also appears IF the CLMI is in a failed state.

Selection of a **CLMI** target is not required prior to selection of a **PORT**. The CLMI is implicitly defined as the board on the same display row as the **PORT** selected.

Note: The **CLM BUILDER** and **CONFIG DATA** targets are now removed from the display, you must be at the CLM Module display to activate these targets – that is, reselect the CLM node.

Continued on next page

1.6 Remote Network Status Display, Continued

PORT

Selection of a specific **PORT** revises the display per Figure 1-6.

RESTART CLMI

Selection of the **RESTART CLMI** requests the system to restart the currently selected Board. The current CLMI is highlighted with a blue background upon selection. The system downloads the board application software to the board from the configured media, place the board in an operational state, then returns control to the display.

Note: This will only appear when the interface board is in the FAIL state.

CHANGE STATE

Selection of **CHANGE STATE**, after selecting either a specific **CLMI** or a specific **PORT**, allows you to change the operating state of the network component. The **IN_SERV** and **OUT_SERV** targets display. The current operating state of the component is shown on the bottom line inside the component's target.

You can change operating state by selecting the appropriate target and then pressing ENTER. Allowable state changes are shown in Table 1.6-1.

Note

The CHANGE STATE target only appears if the station key position matches the minimum access configured on the CLM Configuration Information Page. Figure 1-4a.

Table 1.6-1 State Change Contingency

IF the current state equals ...	THEN you can ONLY change the state to...
OUT_SERV	IN_SERV
IN_SERV	OUT_SERV
FAILED	OUT_SERV

If the requested state change is not allowed per the above table, a message "Invalid selection at this time" appears at the top of the current display.

When a network component's operating state changes to OUT_SERV, all components at lower levels in the network structure have their communications device status changed to PRI_UNAV (Primary Unavailable).

Transition operating states for a network component are shown in Table 1.6-2. These appear on the third line of a component target.

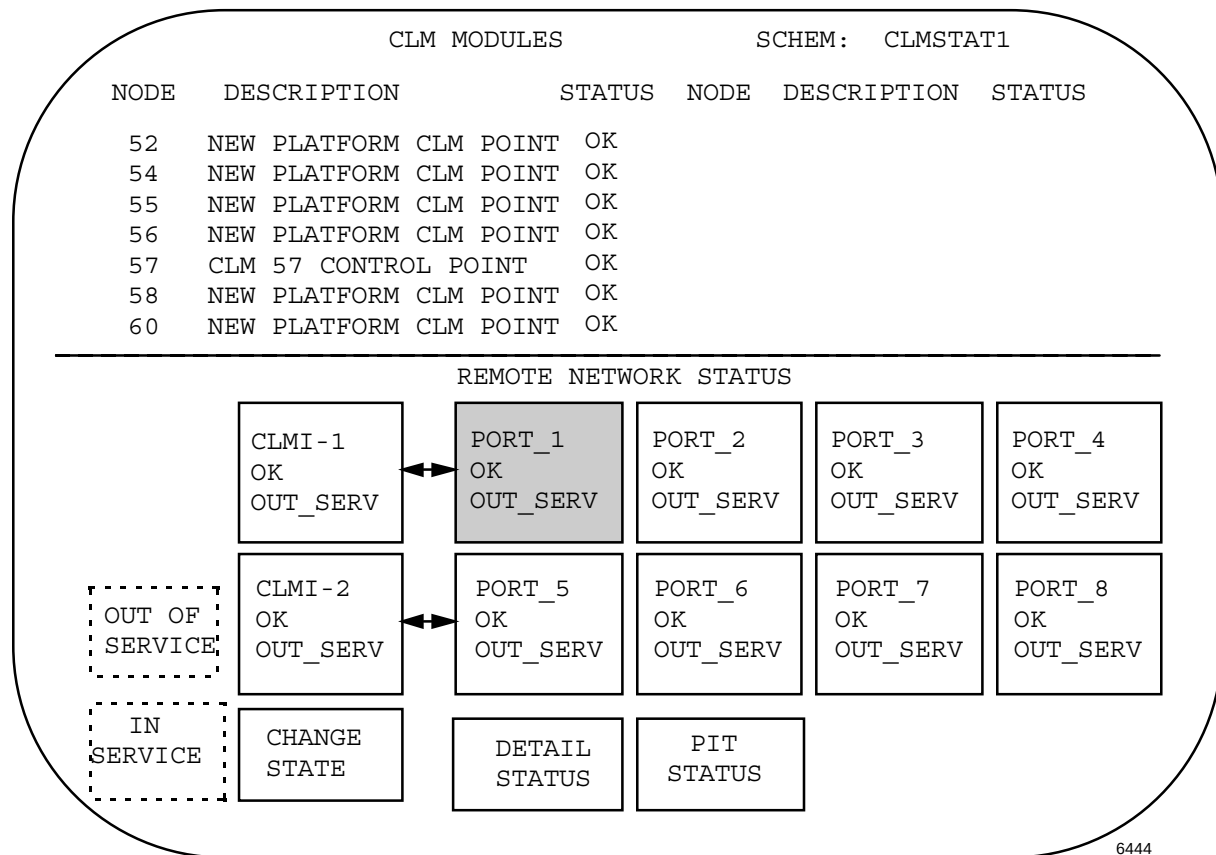
Table 1.6-2 Transition Operating States

STATE	Color	Description
OUT_SERV	Yellow	The component is not to be operated.
SYNC	Yellow	The component is in-service, for data base synchronization only.
IN_SERV	Cyan	The component is in-service, and operating normally.
FAILED	Red	The component has failed. The operator must clear the FAILED state by placing the component out-of-service, and then in-service.

Continued on next page

1.6 Remote Network Status Display, Continued

Figure 1-6 Remote Network Status Display - After Specific PORT selection.



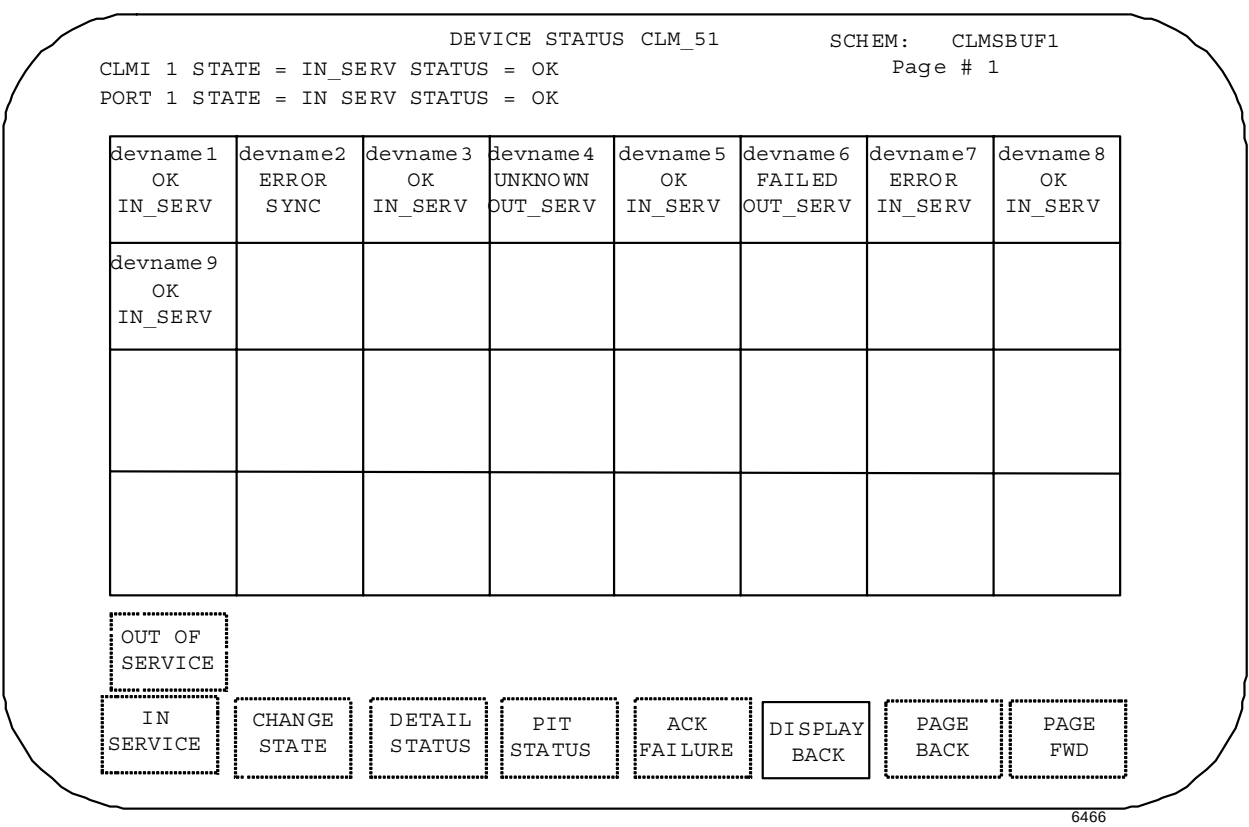
The current port target is highlighted with a blue background upon selection. Three additional targets are shown at the bottom of the display, **DETAIL STATUS**, **PIT STATUS**, and **CHANGE STATE**.

Selecting a specific **CLMI**, or a different **PORT** deactivates the current port selection

- **DETAIL STATUS** transfers to the CLM Network Device Status display, Figure 1-7.
- **PIT STATUS** transfers to the CLM PIT Status display, Figure 1-11.
- **CHANGE STATE** allows changing the operating state of the current port, if the state change access level permits. Ref. Table 1.6-1 and 1.6-2.

1.7 Device Status Display

Figure 1-7 Device Status Display



Note: This may be a multi-page display. If a display page includes the final devices assigned to a port, the remaining device targets display blank.

The Device Status display shows the individual devices on the selected board and port. From this display you can select these targets.

- A specific **DEVICE** which revises the display per Figure 1-8
- The **ACK FAILURE** to acknowledge all devices that are displaying a blinking FAIL or P_FAIL status in red.

Note: This **does not** acknowledge the alarm on the System Process Alarm Display.

The **ACK FAILURE** target is available if there is at least one device on the current display with a communications status equal to FAIL or P_FAIL (blinking in red). The target will not be available after a specific **DEVICE** has been selected. Selection of this target changes all devices on the display that have a blinking FAIL status to ACKFAIL. The literal ACKFAIL displays in red but does not blink. All targets that had a blinking P_FAIL change to a non-blinking P_FAIL.

Note: The **ACK FAILURE** only appears when unacknowledged errors are current on the display.

1.7 Device Status Display, Continued

Device Status Display

From selection of the **DETAIL STATUS** on the Remote Network Status display, Figure 1-6.

The CLM Node name is shown at the top of the display. The selected board and port names, from the previous display, are also shown with their corresponding current operating state and communications status.

The remainder of the display is segmented as follows:

- Up to 5 rows of 8 target boxes, each representing a specific device assigned to the Port. A sixth row at the bottom is reserved for function targets.

Each device target displays the following information for the specific network component:

- Device Name
- Communications Status
- Operating State

Figure 1-8 Device Status Display - After Specific DEVICE Selection

DEVICE STATUS CLM_51 SCHEM: CLMSBUF1
Page # 1

CLMI 1 STATE = IN_SERV STATUS = OK
PORT 1 STATE = IN_SERV STATUS = OK

devname1	devname2	devname3	devname4	devname5	devname6	devname7	devname8
OK IN_SERV	ERROR SYNC	OK IN_SERV	UNKNOWN OUT_SERV	OK IN_SERV	FAILED OUT_SERV	ERROR IN_SERV	OK IN_SERV
devname9 OK IN_SERV							

OUT OF SERVICE

IN SERVICE CHANGE STATE DETAIL STATUS PIT STATUS ACK FAILURE DISPLAY BACK PAGE BACK PAGE FWD

The current **DEVICE** target is highlighted with a blue background.

Communications Status is on the second line of the device target, Ref. Table 1.6-1

Operating State is on the third line of the device target, Ref. Table 1.6-2

- **CHANGE STATE** allows changing the operating state of the current device (if the state change access level permits). Ref. Table 1-4 and 1-5.
- **DETAIL STATUS** transfers to the Device Detail Status display, Figure 1-10.
- **PIT STATUS** transfers to the PIT Status Display, Figure 1-11.

1.7 Device Status Display, Continued

Figure 1-9 CLM Display Navigation Summary, Part 2

The left column contains every display from which you can access another display.

For any given row, the middle two columns define the targets that must be selected in order to show the display in the right column of the same row.

Note: Part 1 of this CLM Display Navigation Summary is shown in Figure 1-2.

IF The Current Display (Figure #nn) Is . . .	AND The Item Selected Is . . .	AND The Function Target Selected Is . . .	THEN The New Display To Be Presented Will Be . . .
Device Status -Figure 1-7	DEVICE		Device Status -Figure 1-8
	DEVICE	DETAIL STATUS _____	Device Detail Status -Figure 1-10 _____
		PIT STATUS	PIT Status - Figure 1-11
PIT Status -Figure 1-11	PIT		PIT Status - Figure 1-12
	PIT	PIT DETAIL _____	PITE Overview -Figure 1-13 _____
		PIT GROUP	PITG -Figure 1-16
PITE Status -Figure 1-14	PT. PARAM		PITE Status -Figure 1-14
PITG - Figure 1-15	PIT	_____	PITG -Figure 1-16 _____
		PIT DETAIL	PITE Overview -Figure 1-13

1.8 CLM Device Detail Status Display

Figure 1-10 Device Detail Status Display

```

CLMI 1 PORT 1                DEVICE DETAIL STATUS                SCHEM:  CLMDETD1
DEVICE NAME devname8                CLM_51

LAST UPLOAD      OK                ADDRESS 1.1.8
LAST DOWNLOAD    OK

xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxx Detailed status information is displayed here...this level of          xxx
xxx information is application and device dependent...the CLM Platform      xxx
xxx provides the mechanism to interface this data...please reference        xxx
xxx the CLM Status Definition File Configuration section of this manual.xxxx

                                o
                                o
                                o

xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx

[UPLOAD] [DOWNLOAD] [DISPLAY] [PAGE] [PAGE]
[DEVICE] [DEVICE]  [BACK]   [BACK] [FWD]

```

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The following targets are specific to a CLM application and do not apply to all applications. Reference the CLM application manual for the device.

- **UPLOAD DEVICE** uploads the current device database
- **DOWNLOAD DEVICE** downloads the most recently saved database to the current device

Continued on next page

1.8 Device Detail Status Display, Continued

Device detail status

The Device Detail Status display is called when the **DETAIL STATUS** is selected on the Device Status display, Figure 1-8.

At the top of the display the following data: CLM Node number, Board and Port numbers, and Device Name.

Status of the last upload and last download and the device address are displayed next.

- this function is application specific and is not available to all applications.

Additional data on display:

LAST UPLOAD STATUS. This represents the status of the last UPLOAD operation for the device. Status values and their corresponding meanings are:

NOT ALLOWED Not supported for this application.

OK Last upload completed successfully.

FA Last upload operation failed.

IP An upload operation is in-progress.

This target is available if the current device is upline loadable, and an upload/download operation is not currently in progress for the device.

Selection of the **UPLOAD DEVICE** requests the system to upload the current device database to an HM file. The file name is assigned automatically by the system to the targeted device.

LAST DOWNLOAD STATUS. This represents the status of the last DOWNLOAD operation for the device. Status values, and their corresponding meanings, are:

NOT ALLOWED Not supported for this application.

OK Last download completed successfully.

FA Last download operation failed.

IP A download operation is in-progress.

This target is available if the current device is downline loadable, and an upload/download operation is not currently in-progress for the device.

Selection of the **DOWNLOAD DEVICE** requests the system to download the database from an HM file to the current device.

The system retrieves the HM file that contains the most recently uploaded database for the specific device. Therefore, at least one upload operation must have been completed successfully for this device before target can be activated

DEVICE ADDRESS. This represents the address of the device in the network. The address format is digit(s), separated by a period, for each level in the network hierarchy (i.e., "1.2.11.234"). The network hierarchy starts at the Board, then Port, then each addressable component down to, and including, the current network device.

Detailed operating status information

The remainder of the display consists of detailed operating status information for the selected device. Specific literals/codes that portray this detailed status information are application and device dependent. Refer to the specific device CLM application manual.

Continued on next page

1.9 CLM Point Interface Table Status Display

Figure 1-11 CLM Point Interface Table (PIT) Status Display

CLM POINT INTERFACE TABLES (PIT) STATUS

SCHEM: CLMPIT1

CLMI (1) PORT (1) CLM_51

PIT NAME	PIT GROUP	DEVICE	BEHIND COUNT	ERROR/COUNT D.A.	DEV.	CYCLE
XXXXXXXX	XXXXXXXX	XXXXXXXX	0	1	0	5 sec
XXXXXXXX	XXXXXXXX	XXXXXXXX	3	2	5	5 sec
XXXXXXXX	XXXXXXXX	XXXXXXXX	0	0	0	10 sec
---	---	---	---	---	---	---
---	---	---	---	---	---	---
---	---	---	---	---	---	---
XXXXXXXX	XXXXXXXX	XXXXXXXX	2	0	3	10 sec
XXXXXXXX	XXXXXXXX	XXXXXXXX	0	1	2	5 min
XXXXXXXX	XXXXXXXX	XXXXXXXX	0	4	6	15 min

CHANGE CYCLE

DEMAND PROCESS

PIT DETAIL

PIT GROUP

DISPLAY BACK

PAGE BACK

PAGE FWD

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The following targets are initially available from the PIT Status display:

- A specific **PIT NAME** line target is the entire line, which revises the display per Figure 1-12.

PIT Status Display

Called from:

- The Remote Network Status display, Figure 1-6. Displays the PITs related to the current selected Board or Port.
- The Device Status display, Figure 1-8. Displays the PITs related to the current selected Device.

Continued on next page

1.9 CLM Point Interface Table Status Display, Continued

Data line information

PIT Status Display Data Line Information

Item	Explanation
PIT Name	PIT Name: The name of the PIT represented in this position.
PIT Group	PIT Group Name: If the PIT is a member of a PITG, the name of the PITG is shown; otherwise, this is blank.
Device	Device Name: The name of the device where the actual data points within the PIT are located.
Behind Count	Process Counter: A count of the number of cycles behind, if any, while attempting to process the PIT at the defined processing frequency.
Error Count/D.A.	Data Access Error Count: The total number of LCN Data Access errors encountered, during the last process cycle of the PIT (Stores/Reads from LCN point.parameter).
Error Count/Dev.	Device Access Error Count: The total number of Device data access type errors encountered, during the last process cycle of the PIT.
Cycle	Cycle Time: This value represents the time duration, or the time increment, between scheduled processing of the PIT. Also known as processing frequency. Meanings for the time-unit suffix are: • SEC Seconds (2 – n) • MIN Minutes (1 – n) • HR Hour(s) (1 – 4095) • EXC Exclusive: A value of 0 (zero) is defined as the exclusive scan, which is a priority fast scan for this PIT only • N/S Not Scheduled (-1)

Continued on next page

1.9 CLM Point Interface Table Status Display, Continued

Figure 1-12 CLM PIT Status Display - After specific PIT Name line selection

CLM POINT INTERFACE TABLES (PIT) STATUS SCHEM: CLMPIT1
 CLMI (1) PORT (1) CLM_51

PIT NAME	PIT GROUP	DEVICE	BEHIND COUNT	ERROR/COUNT		
				D.A.	DEV.	CYCLE
xxxxxxxx	xxxxxxxx	xxxxxxxx	0	1	0	5 sec
xxxxxxxx	xxxxxxxx	xxxxxxxx	3	2	5	5 sec
xxxxxxxx	xxxxxxxx	xxxxxxxx	0	0	0	10 sec
_____	_____	_____	_____	_____	_____	_____
_____	_____	_____	_____	_____	_____	_____
xxxxxxxx	xxxxxxxx	xxxxxxxx	2	0	3	10 sec
xxxxxxxx	xxxxxxxx	xxxxxxxx	0	1	2	5 min
xxxxxxxx	xxxxxxxx	xxxxxxxx	0	4	6	15 min

CHANGE
CYCLE

DEMAND
PROCESS

PIT
DETAIL

PIT
GROUP

DISPLAY
BACK

PAGE
BACK

PAGE
FWD

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Select PIT NAME target

The current PIT line is highlighted with a blue background upon selection. Additional targets are shown at the bottom of the display.

If the current PIT selected is a member of a PIT Group the **PIT GROUP** target displays. If Keyswitch = ENGR the **CHANGE CYCLE** target displays.

- **PIT DETAIL** transfers to the CLM PITE Overview display, Figure 1-13.
- **PIT GROUP** transfers to the CLM PITG display, Figure 1-15.

Select CHANGE CYCLE

This target is available if the keyswitch is in the engineer position.

Allows a change of the processing frequency for the selected PIT. Cycle represents the time duration between scheduled processing of the PIT. Enter an integer between 1 and 4096, then enter the corresponding suffix: 1 = Seconds; 2 = Minutes; 3 = Hours. A '-1' can be entered to stop processing of the PIT.

Note: Two seconds is the fastest schedule time currently allowed. The value 0 (zero) can be entered to enable a priority fast scan for the PIT; this exclusive scan is performed to the exclusion of all other periodic scans on a network Port. The exclusive scan inhibits input of slow cycle processing. The value of -1 inhibits scheduling of the PIT. However, the PIT can still be scheduled through a PIT Group.

Continued on next page

1.9 CLM Point Interface Table Status Display, Continued

Select DEMAND PROCESS

Requests the system to initiate a one-time immediate process of the selected PIT. While this request is in-progress the **DEMAND PROCESS** displays in white, and all other targets at the bottom of the display are removed. Upon completion, the display refreshes per Figure 1-12.

1.10 CLM PITE Overview Displays

Figure 1-13 CLM Point Interface Table Entry Overview Display

CLM POINT INTERFACE TABLE (PITE) STATUS SCHEM: CLMLIST1
CLMI(1) PORT(1) CLM_51 Page No. 1
POINT INTERFACE TABLE NAME = xxxxxxxx
PIT ENTRY OVERVIEW

xxxxxxxxxxxxxxxxxxxxxx	xxxxxxxxxxxxxxxxxxxxxx	xxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxxxx	xxxxxxxxxxxxxxxxxxxxxx	xxxxxxxxxxxxxxxxxxxxxx
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
xxxxxxxxxxxxxxxxxxxxxx	xxxxxxxxxxxxxxxxxxxxxx	xxxxxxxxxxxxxxxxxxxxxx
xxxxxxxxxxxxxxxxxxxxxx	xxxxxxxxxxxxxxxxxxxxxx	xxxxxxxxxxxxxxxxxxxxxx

PITE DETAIL DISPLAY BACK PAGE BACK PAGE FWD

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PITE overview

Called from:

- CLM PIT Status display, Figure 1-12.
- CLM PITG display, Figure 1-15.

Displays the list of image points that are related to the previously selected PIT.

Data displayed

The currently selected CLM node, board, port, and PIT values are all shown at the top of the display. The display lists all image points associated with PITEs in this PIT. This is a multipage display with a maximum of 30 image points listed per page.

Error indications

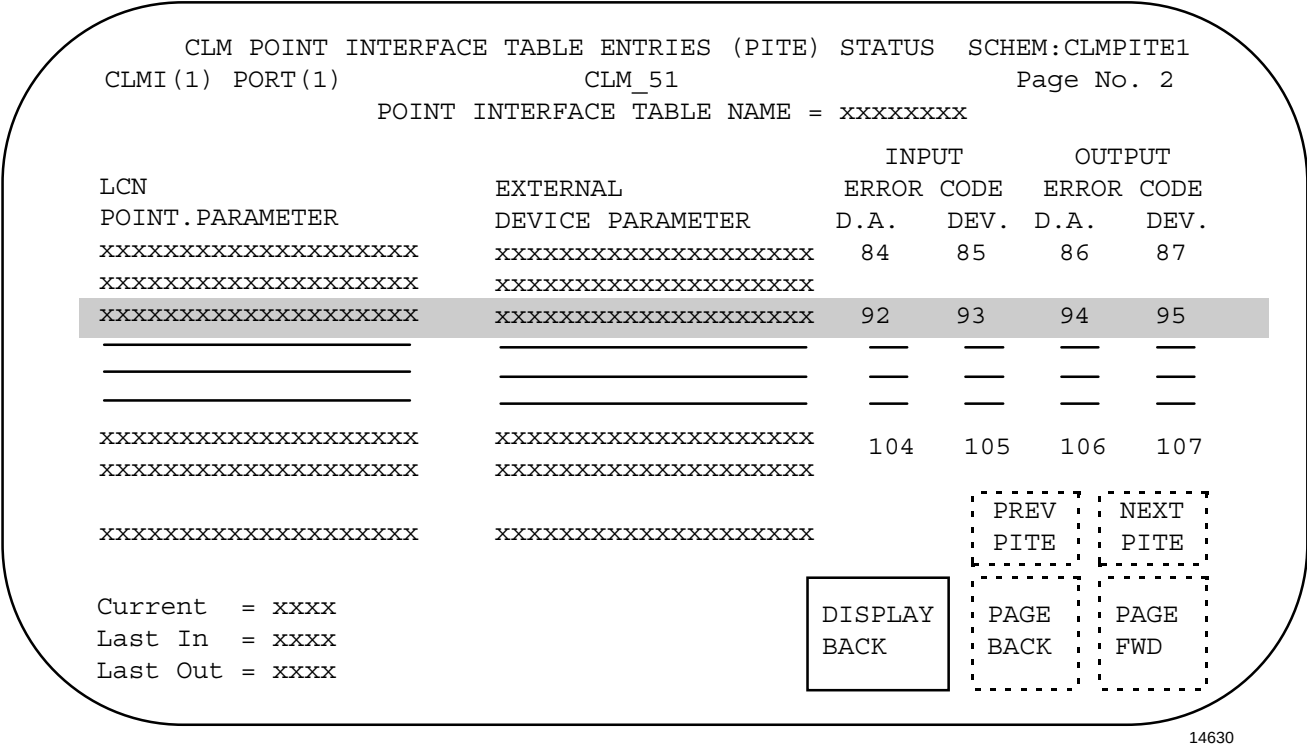
The normal text color for the image point list is cyan to represent no input or output errors specific to each entry. If errors do exist, the text changes to yellow. To view the specific errors for an entry, select the entry and then the **PITE DETAIL** to call the PITE Status display, Figure 1-14.

Select PITE DETAIL

Selection of the **PITE DETAIL**, after selecting a specific PITE image point, transfers the user to the PITE Status display, Figure 1-14.

1.11 CLM Point Interface Table Entry Status Display

Figure 1-14 CLM Point Interface Table Entry (PITE) Status Display



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Note: This display may be a multipage display.

PITE Status Display

Called from:

- CLM PITE Overview display, Figure 1-13.

PITE targets

The following targets are initially available from the PITE Status display:

- A specific PITE or Point.Parameter Name/line. This will result in the display of the current, last in, and last out values for the selected point/parameter. This may be either numeric or character data depending on the selected parameter. Also, depending on the point type, all values may not be used. For example, An Analog Input point would not require the Last Out value.
- A specific PITE Data Access or Device error code on the Point.Parameter line. This will result in the display of 16 character descriptors for the input and output error codes for the Data Access and Device errors. If the error code is not recognized, it will show “UNDEFINED ERROR”.
- Select **PREV PITE**, if displayed, to display the prior PITE in the list from the previous PITE Overview display
- Select **NEXT PITE**, if displayed, to display the next PITE in the list from the previous PITE Overview display

1.11 CLM Point Interface Table Entry Status Display, Continued

Data displayed

The currently selected CLM Node, Board, Port, and PIT values are shown at the top of the display.

Data line information

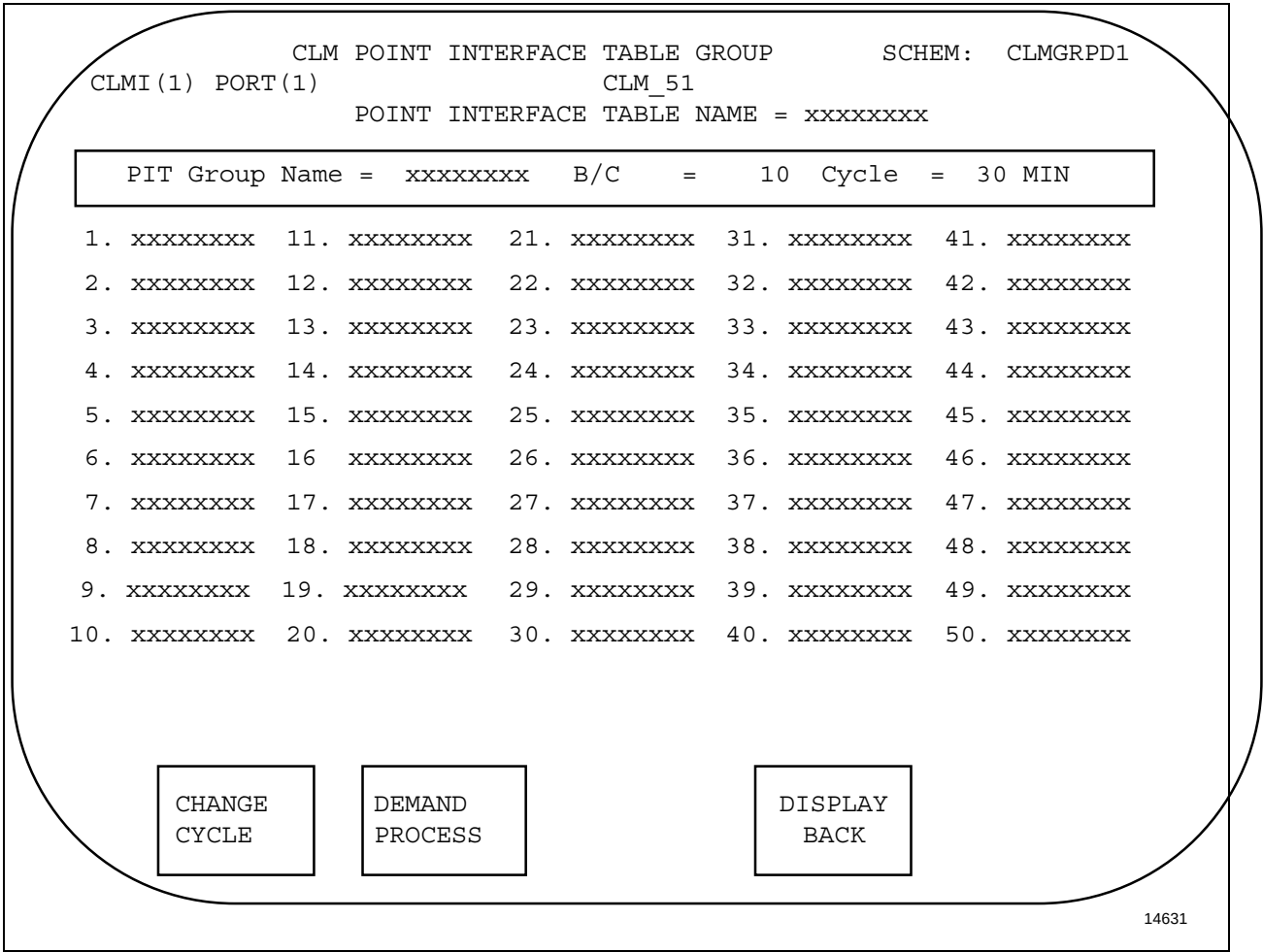
PITE Status Display Data Line Information

Item	Explanation	
PITE name	Point.Parameter Name: The name of the LCN image point and associated parameter.	
External name	External Device Parameter Name or address: The name or address of the point.parameter in the non-Honeywell device, if this device uses point.parameter names.	
Input error codes	D.A. Data Access (DA) Status Code. Last data access status code for an attempt to store to the LCN input parameter. DA codes are defined in the <i>Messages Directory</i> in the TDC 2030-1 binder.	
	DEV. Device Access Status Code. Last device access status code for input parameter. This may vary by protocol.	
	DEV. Status Code	Meaning
	(Blank)	Okay
	1 or > 3	Error, Reference application specific error codes. MODBUS – Appendix A.8 and A.9 BSAP – Appendix A.4 – A.7
	2	Failed
	3	Unknown
Output error codes	D.A. Data Access Status Code. Last data access status code for an attempt to read from the LCN output parameter. (Definition: same as above.)	
	DEV. Device Access Status Code. Last device status code for output parameter. (Definition: same as above.)	

Continued on next page

1.12 CLM Point Interface Table Group Display

Figure 1-15 CLM Point Interface Table Group (PITG) Display



PITG display

Called from:
PIT GROUP target on the CLM PIT Status display, Figure 1-12.

The currently selected CLM Node, board, and port are shown at the top of the display. The display highlights the PIT Group that the previously selected PIT is a member of with the PITG's Behind Count and Cycle data.

The remainder of the display lists all PITs that are members of the highlighted PIT Group. Because the maximum of 50 PITs allowed in a PITG are all shown on the display, paging is not required.

1.12 CLM Point Interface Table Group Display, Continued

PIT Group Name

Selection of a specific PIT Name revises the display per Figure 1-16.

PIT DETAIL also displays as an option. If keyswitch = ENGR **CHANGE CYCLE** displays.

CHANGE CYCLE

This target is available with the keyswitch in the Engineer position.

Selection of the **CHANGE CYCLE**, after selecting a specific PIT Name or the PIT Group, allows you to change the processing frequency for the selected PIT or PITG. Cycle represents the time duration between scheduled processing of the PIT or PITG. Enter an integer between 1 and 4096 and the corresponding suffix: 1 = Seconds; 2 = Minutes; 3 = Hours.

Note: Two seconds is the fastest schedule time currently allowed. The value 0 (zero) can be entered to enable a priority fast scan for the PIT; this exclusive scan is performed to the exclusion of all other periodic scans on a network Port. The exclusive scan inhibits input of slow cycle processing. The value of -1 inhibits scheduling of the PIT. However, the PIT can still be scheduled through a PIT Group.

DEMAND PROCESS

Selection of the **DEMAND PROCESS**, after selecting a specific PIT Name or the PIT Group, requests the system to initiate a one-time immediate process of the selected PIT or PITG. While this request is in progress, the **DEMAND PROCESS** is outlined in white, and all other targets at the bottom of the display are removed. Upon completion, the display refreshes per Figure 1-16.

PIT DETAIL

Selection of **PIT DETAIL**, after selecting a specific PIT Name, transfers you to the CLM PITE Overview display, Figure 1-13. This display shows the LCN points associated with the current PIT selection.

Continued on next page

1.12 CLM Point Interface Table Group Display, Continued

Figure 1-16 CLM PITG Display - After Specific PIT Name Selection

CLM POINT INTERFACE TABLE GROUP

SCHEM: CLMGRPD1

CLMI (1) PORT (1)

CLM_51

POINT INTERFACE TABLE NAME = xxxxxxxx

PIT Group Name = xxxxxxxx

B/C = 10

Cycle = 30 MIN

1. xxxxxxxx

11. xxxxxxxx

21. xxxxxxxx

31. xxxxxxxx

41. xxxxxxxx

2. xxxxxxxx

12. xxxxxxxx

22. xxxxxxxx

32. xxxxxxxx

42. xxxxxxxx

3. xxxxxxxx

13. xxxxxxxx

23. xxxxxxxx

33. xxxxxxxx

43. xxxxxxxx

4. xxxxxxxx

14. xxxxxxxx

24. xxxxxxxx

34. xxxxxxxx

44. xxxxxxxx

5. xxxxxxxx

15. xxxxxxxx

25. xxxxxxxx

35. xxxxxxxx

45. xxxxxxxx

6. xxxxxxxx

16. xxxxxxxx

26. xxxxxxxx

36. xxxxxxxx

46. xxxxxxxx

7. xxxxxxxx

17. xxxxxxxx

27. xxxxxxxx

37. xxxxxxxx

47. xxxxxxxx

8. xxxxxxxx

18. xxxxxxxx

28. xxxxxxxx

38. xxxxxxxx

48. xxxxxxxx

9. xxxxxxxx

19. xxxxxxxx

29. xxxxxxxx

39. xxxxxxxx

49. xxxxxxxx

10.

xxxxxxx

20. xxxxxxxx

30. xxxxxxxx

40. xxxxxxxx

50. xxxxxxxx

B/C

D/A

DEV

CYCLE

10

2

168

20 MIN

CHANGE CYCLE

DEMAND PROCESS

PIT DETAIL

DISPLAY BACK

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The following is a description of the additional data fields shown on Figure 1-16:

- B/C** Behind Count/Process Counter: A count of the number of cycles, if any, that the system is currently behind while attempting to process the PIT or PITG at the defined processing frequency.
- D/A** Data Access Error Count: The total number of LCN Data Access errors encountered, during the last process cycle of the PIT or PITG.
- DEV** Device Access Error Count: The total number of Device data access type errors encountered during the last process cycle of the PIT or PITG.
- Cycle** Cycle Time: The minimum cycle is two seconds. Represents the time duration, or the time increment, between scheduled processing of the PITG. Also known as processing frequency. The time-unit suffixes are:
 - SEC** = Seconds; **MIN** = Minutes; **HR** = Hours.
 - NS** = Not scheduled, value of -1.
 - EXC** = Exclusive. A value of 0 (zero) is defined as the exclusive scan, which is a priority fast scan for this PIT or PITG only.

Section 2 – User Functions

2.1 Introduction

The following sections describe specific steps required to perform user functions provided with the CLM Platform displays.

All of the display figures referenced are located in the previous section: CLM Normal Operating Environment.

2.2 Change Cycle of a PIT or PITG

Change Cycle PIT or PITG

Note: To change cycle, the keyswitch must be in the **Engineer** position.

The **CHANGE CYCLE** target allows you to change the processing frequency for the selected PIT or PITG. Cycle represents the time duration between scheduled processing of the PIT or PITG. Enter an integer between 1 and 4096, then enter the corresponding suffix: 1 = Seconds; 2 = Minutes; 3 = Hours. The minimum cycle is two seconds.

Note: The value 0 (zero) can be entered to enable a priority fast scan for the PIT or PITG. This exclusive scan is performed to the exclusion of all other periodic scans on a network Port and slow scans. The value of -1 can be entered to inhibit scheduling for the PIT or PITG. If a PIT is assigned to a PITG, setting the cycle time to -1 does not change the PITG scheduling of the PIT.

To Change Cycle of a PIT or PITG

Step	Display / Fig. No.	User Action
1	CLM Modules, Fig. 1-3	Bring Up CLM Modules display
2		Select desired CLM Node.
3	Remote Network Status, Fig. 1-4	Select the desired PORT , which represents the Board and Port within the desired PIT or PITG address hierarchy.
4		Select PIT STATUS .
5	PIT Status, Fig. 1-11	Select specific PIT Name line target that contains the desired: • PIT Name (in first column), or • PIT Group Name (in second column). Use PAGE FWD if necessary.
Note: Perform Step 6 for PIT OR steps 7- 8 for PITG.		
6	PIT Status, Fig. 1-12	To Change Cycle of a PIT: Select CHANGE CYCLE then respond to the prompts for cycle value and cycle time unit.
7		To Change Cycle of a PITG: Select PIT GROUP after selecting the PIT Name line target that contains the desired PITG in second column.
8	PIT Group, Fig. 1-16	Select CHANGE CYCLE then respond to the prompts for cycle value and cycle time unit.
Note: Make sure the desired PITG is still highlighted on the display (If a PIT is selected on this display, the Change Cycle is applicable to the PIT, rather than the desired PITG.).		

Continued on next page

2.3 Change State: Board, Port, or Device

Change State

Note: On the CLM Configuration Information page verify that the access level and the keyswitch position allow for the **CHANGE STATE** target to be visible.

The current operating state of the network component is shown on the bottom line inside the component's target box.

When **CHANGE STATE** is selected, the **IN_SERV** and **OUT_SERV** targets display.

To change the operating state, select the appropriate target and then press ENTER.

Allowable State Change Conditions

IF the current state equals ...	THEN you can ONLY change to...
OUT_SERV	IN_SERV
IN_SERV	OUT_SERV
FAILED	OUT_SERV

If the requested state change is not allowed, per this table, the message "INVALID SELECTION AT THIS TIME" appears at the top of the display.

Change State

Step	Display / Fig. No.	User Action
1	CLM Modules, Fig. 1-3	• Bring Up CLM Network Status Display • Select CLM Node
Note: Perform Step 2,3 OR 4 for CLMI, Port, OR Device Respectively.		
2	Remote Network Status, Fig. 1-4	• Select specific component: Board, Port, or Device • Select CHANGE STATE • Select IN_SERV or OUT_SERV • Press the ENTER key

Continued on next page

2.4 Checkpoint the CLM

CLM Checkpoint

The CLM data base is checkpointed whenever the Application Module SAVE DATA is performed. This CLM checkpoint function is automatic when this action is performed and no other specific user interaction is required.

The last successful checkpoint time/date is shown at the bottom of the Remote Network Status display. If the time/date text color is cyan, then the checkpoint is current. Yellow text and a yellow asterisk by the CLM node number indicate that a checkpoint has not yet been performed after CLM Builder changes were made. Red text and asterisk indicate that a CLM checkpoint was attempted but was not successful. Check journal messages for specifics.

Note: Due to interaction between the AM checkpoint and CLM checkpoint, several 'completion' messages may be journalled. This duplication may be disregarded.

CLM Data Destination Path

When an AM SAVE DATA is performed, the CLM data is also saved. The CLM Platform tasks create a file in the format:

CLM nnn .CP

where nnn = the three digit node number (i.e. 049)

This file is created and copied to the destination pathname defined in the PATHNAME CONFIGURATION for SAVE DATA on the CLM CONFIGURATION Page.

Note that the directory only is specified, since the logical device (i.e. HM, \$F1, etc) specified by the AM SAVE DATA operation will be used.

2.5 Demand Process: PIT or PITG

Demand Process

Demand Process immediately updates all input and output cyclic data within the selected PIT or PITG.

While the request is in-progress the **DEMAND PROCESS** displays in white and all other targets at the bottom of the display are removed. Upon completion, the display refreshes.

To Demand Process PIT or PITG

Step	Display / Fig. No.	User Action
1	CLM Modules, Fig. 1-3 Remote Network Status, Fig. 1-4 Fig. 1-6 Fig. 1-11	• Call CLM Modules display, CLMSCHEM. • Select the CLM Node. • Select specific PORT , which represents the Board and Port within the desired PIT or PITG address hierarchy. • Select PIT STATUS . • Select specific PIT Name line target that contains the desired PIT or PIT Group Name. PAGE FWD until the desired PIT Name or PIT Group Name is displayed.
Note: Perform Step 2 OR 3 (for a PIT OR a PITG Respectively).		
2	PIT Status, Fig. 1-12	To Demand Process a PIT: • Select DEMAND PROCESS • Press the ENTER key.
3	Fig. 1-12 PIT Group, Fig. 1-16	To Demand Process a PITG: • Select PIT GROUP after selecting the PIT Name line target that contains the desired PITG in second column. • Select DEMAND PROCESS • Press the ENTER key. • Note: Make sure the desired PIT Group is still highlighted on the display (if a PIT is selected on this display, the Demand Process is applicable to the PIT, rather than the desired PITG).

Continued on next page

2.6 Upload or Download a Specific Device

The Upload and Download function is specific to a CLM application and is not required or supported for most applications. Refer to the specific CLM application manual for the device to determine if this function is supported.

Upload Device

This target is available if the current device is upline loadable and an upload/download operation is not currently in-progress for the device.

Selection of **UPLOAD DEVICE** requests the system to upload the current device database to a file on the configured media. The file name is assigned automatically by the system to the targeted device.

Download Device

This target is available if the current device is downline loadable and an upload/download operation is not currently in-progress for the device.

Selection of **DOWNLOAD DEVICE** requests the system to download the database from a file on the configured media to the current device. The system retrieves the file that contains the most recent uploaded database for the specific device. Therefore, at least one upload operation must have been performed successfully for this device before the **DOWNLOAD DEVICE** function can be performed.

To Upload or Download a Device

Step	Display / Fig. No.	User Action
1	CLM Modules, Fig. 1-3 Remote Network Status, Fig. 1-4 Fig. 1-6 Network Device Status, Fig. 1-7 Fig. 1-8 Device Detail Status, Fig. 1-10	• Call CLM Modules Display, CLMSCHEM. • Select the CLM Node. • Select specific PORT , which represents the Board and Port within the desired device address hierarchy. • Select DETAIL STATUS . • Select specific DEVICE . • Select PAGE FWD until desired device is on display. • Select DETAIL STATUS . • Select the desired target: • UPLOAD DEVICE • DOWNLOAD DEVICE

Continued on next page

2.7 Restart the Interface Board

Note: This target only appears when the CLMI, CCP, or UCIM Interface board is in the FAILED status.

Restart CLMI target

Selection of **RESTART CLMI** requests the system to restart the currently selected board. The current CLMI target is highlighted with a blue background upon selection. The system downloads the board and application software from the configured media to the board, places it in an operational state, and then returns control to the display.

To Restart the Interface Board

Step	Display / Fig. No.	User Action
1	CLM Modules, Fig. 1-3 Remote Network Status, Fig. 1-4	• Call CLM Modules Display, CLMSCHEM. • Select the CLM Node • Select specific CLMI • Select RESTART CLMI • Press the ENTER key

2.8 Respond to Device Failure

Display / Fig. No.	Failure Alarm
Remote Network Status, Fig. 1-4	PORT target blinking, red FAIL or P_FAIL.
Network Device Status, Fig. 1-7	DEVICE target blinking, red FAIL or P_FAIL

To Respond to a Device Failure

Step	Display / Fig. No.	User Action
1	Remote Network Status, Fig. 1-4 Fig. 1-6 Device Status, Fig. 1-7	- Select blinking, red PORT target. This represents the board and port within the failed device address hierarchy. The selection revises targets at the bottom of the display. - Select DETAIL STATUS. - Select ACK FAILURE to acknowledge all devices that are displaying a blinking FAIL or P_FAIL status in red. Note: After selecting ACK FAILURE, any failed devices change from blinking FAIL status in red to ACKFAIL status in red (not blinking). Any devices that are in the partially failed status change from blinking P_FAIL to non-blinking P_FAIL. - Select specific DEVICE. Note: PAGE FWD to desired device
2	Network Device Status. Fig. 1-8 Device Detail Status, Fig. 1-10	To Review Failed Device Detailed Status: - Select DETAIL STATUS - Review detailed operating status information for the device, then - Select DISPLAY BACK to return to Network Device Status with the same device selected.
3	Network Device Status, Fig. 1-8	To Restore Device from FAIL/ACKFAIL State: - Select CHANGE STATE - Select OUT_SERV - Select CHANGE STATE - Select IN_SERV
Important: Step 3 applies to devices that are hard-failed (i.e., STATE=FAIL). Devices that have the status FAILED attempt to recover automatically.		

Continued on next page

Appendix A – CLM Error Messages

A.1 CLM Tasks in the AM Personality

Task numbers and descriptions

The CLM platform package in the AM side (i.e. not board application software) consists of nine tasks that run independently within the normal Application Module tasking hierarchy. These tasks are overlaid onto the Custom Tasks (CUST_TASK_nn) that are available in every AM. The AM assigns these tasks as they are created to the next available custom task slot in the list. If the data listed on page 2 of the LCN node's NCF data for the CLM node indicates that there are Background CL tasks and Concurrent data Access tasks assigned to the node (minimum of 2 and 4 respectively), then these are assigned first to the custom task list before the CLM tasks are assigned. Care must be exercised to not exceed the total number of tasks available in the AM (See note below).

CLM Task Numbers and Descriptions

Task Number*	Task Description
n+1	Startup Task and CLM Operator Display Task
n+2	Scheduler—initiates the transfer of information to/from the CLM database and the Networked Devices.
n+3	Synchronization Processor—keeps the CLM database synchronized with the databases of the Networked Devices.
n+4	Fetcher—gets values from the TPN database via Data Access and places them in the CLM Network Memory database.
n+5	Storer—puts values from the CLM Network Memory database into the TPN database using Data Access.
n+6	Message Handler—transfers messages to/from the message router on the interface board. Most of the CLM tasks communicate with interface board functions in this manner.
n+7	Intercepts—Interprets all operator and CL-based stores in the local CLM node; puts values into the CLM Network Memory database for output to the Networked Devices.
n+8	Builder—creates and modifies the CLM Network Memory Database. This database is not part of the TPN database. This task also processes the save of Network Memory data.
n+9	CLM Database Interface—routines that access/update the CLM Database.

***Note:** The CLM tasks run within the custom task set available in the AM. R430 has 25 tasks available, and R500 and later have 30 tasks available. The n referenced under the task number includes the number of background CL and Concurrent Data Access tasks (Refer to Section 4.4 of the *CLM Platform Implementation Manual*). Each additional task configured increments the task numbers by one (for example, seven background CL tasks and four Concurrent Data Access tasks will cause the task numbers to range from 12 through 20).

A.2 CLM Platform Error Codes

Following is a list of error codes for the CLM Platform. These codes appear as the first status value [that is, Int(1) in "System Errors"] logged in the TPN System Error Journal.

There are four status values logged in each System Error message. The status values are referred to as Int(1) through Int(4) in the descriptions below. Int(1) is the left most value in the message and Int(4) is the right most. The error code is always Int(1).

The descriptive error text is also displayed in the CLM MODULE Status object on the Status Detail summary page and the Status Notification Journal for the CLM node.

100-199 – Startup task	101	Load_Netmem_Chkpt unable to Get_FM_Request_Block
	102	Load_Netmem_Chkpt unable to Open_File
	103	Load_CLMI_Module unable to Get_FMRB for Object File
	104	Load_CLMI_Module unable to open Object File
	105	Load_CLMI_Module found Network Memory uninitialized
	106	Load_CLMI_Module unable to Get_FMRB for Configuration File
	107	Load_CLMI_Module unable to open Configuration File
	108	Load_CLMI_Module unable to GetMemory for ldb
	109	Load_CLMI_Module found an invalid configuration file
	110	Reset_And_Test_CLMI_Board was given an invalid slot number
	111	Reset_And_Test_CLMI_Board was unable to get board to successfully pass self-test
	112	Reset_And_Test_CLMI_Board was unable to send a "CLMI_ID" command to the specified CLMI board
	113	Reset_And_Test_CLMI_Board did not see a command completion acknowledgment within one second of sending the ID command to the specified CLMI board
	115	Error sending LOAD command to CLMI board Int(2) is Load_CLMI_Module return status Int(3) is the record number in the s-record file that precipitated the error
	116	Error loading s-records into CLMI board Int(2) is Load_CLMI_Module return status Int(3) is the record number in the s-record file that precipitated the error Int(4) is the CLMI Loader status
	117	Error loading s-records into CLMI board. Load is not complete and s-records have been exhausted Int(2) is Load_CLMI_Module return status Int(3) is the record number in the s-record file that precipitated the error Int(4) is the CLMI Loader status
	118	Error processing INIT directive Int(2) is Load_CLMI_Module return status Int(3) length of time in 100MS increments the CLM waited for the CLMI Loader Int(4) is the CLMI Loader status

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A.2 CLM Platform Error Codes, Continued

100-199 – Startup task, continued

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|-----|--|
| 119 | Error processing RUN directive
Int(2) is Load_CLMI_Module return status
Int(3) length of time in 100MS increments the CLM waited for the CLMI Loader
Int(4) is the CLMI Loader status |
| 120 | Error sending GO command to CLMI board
Int(2) is Load_CLMI_Module return status
Int(3) length of time in 100MS increments the CLM waited for the Loader to complete |
| 121 | Init_Netmem found a bad PITE parameter in network memory |
| 122 | Init_Netmem found a bad PITE in network memory |
| 123 | Init_Netmem got an error attempting to Get a PITE parameter |
| 124 | Init_Netmem found a bad PIT/PITG in the PIT Directory |
| 125 | Init_Netmem found a bad starting PITE in a PIT |
| 126 | Init_Netmem found a bad PITE in a PIT |
| 127 | Init_Netmem found a bad NDT entry |
| 128 | Init_Netmem found a bad starting NDT entry in the NDT |
| 129 | Init_Netmem found a bad NDT entry in the NDT |
| 130 | Init_Netmem found a bad starting PIT/PITG in the PIT Directory |
| 131 | Init_Netmem found a bad PIT/PITG entry in the PIT Directory |
| 133 | SAVE_NETMEM_CHKPT failed. Error occurred or not enough memory. |
| 135 | Was unable to startup all of the CLM tasks.
Int(3) is the status returned by Activate_Tasks |
| 136 | Unable to set the Task Priority of the Startup and Display task.
Int(3) is the status returned by \$Set_Current_Priority. |
| 137 | Startup task was unable to allocate the checkpoint semaphore.
Int(2) is the return status from \$allocate_semaphore. |
| 138 | SAVE_NETMEM_CHKPT routine was unable to get the checkpoint semaphore.
Int(2) is the return status from \$get_semaphore. |
| 139 | SAVE_NETMEM_CHKPT was unable to release the checkpoint semaphore.
Int(2) is the return status from \$release_semaphore |
| 140 | Unable to access the ENTARR(101) parameter on the Builder point that is required for the Checkpoint default pathname. ORD(3) = minor; ORD(4) = pa sts. |
| 141 | Error parsing default CP pathname. Invalid pathname parsed. |
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A.2 CLM Platform Error Codes, Continued

200-299 – Builder	200	The Builder was unable to get memory required during a processing sequence.
	201	The Builder was unable to get an FMRB to use to obtain all of the template files on the HM.
	203	The Builder was unable to clear the CDS data arrays on the Builder point after the user left the display.
	204	The Builder was unable to fetch the BUILDER parameter from the Builder point.
	205	The Builder was unable to convert one or more of the required builder point CDS parameters from external to internal format.
	206	The Builder was unable to convert the required Builder point id from external to internal format.
	207	The user has left the Builder display while a requested action was in progress.
	208	The Builder was unable to \$synchronize delay during looping to check for possible builder action.
	209	The Builder received an illegal action code from the Builder display.
	210	The Builder was unable to fetch the ACTION parameter from the Builder point.
	211	The Builder was unable to fetch the LOCK parameter from the Builder point.
	212	Error attempting to checkpoint network memory. Int(3) is the return status from SAVE_NETMEM_CHKPt.
	214	The Builder was unable to get an FMRB for the Template File(TDF).
	215	The Builder was unable to store to the LOCK CDS on the Builder point.
	216	The Builder was unable to store to the VERROR CDS on the Builder point.
	217	The Builder was unable to store to the STATUS CDS on the Builder point.
	218	The Builder was unable to store to the ENTRY CDS on the Builder point.
	221	The Builder was unable to get/put a CDS array because of an IDB related error.
	222	The Builder was unable to clear the DEFARR, VALARR, FMTARR, or STSARR CDS arrays on the Builder point due to an IDB-related error.
	223	The Builder was unable to fetch the ENTRYNAM CDS parameter from the Builder point.
	224	The Builder was unable to store to the VALARR CDS array on the Builder point due to an IDB-related error.
	225	The Builder was unable to store to the FMTARR CDS array on the Builder point due to an IDB-related error.
	227	The Builder was unable to store to the DEFNUM CDS on the Builder point.
	228	The Builder was unable to \$synchronize delay while waiting on an entry to be retrieved that is in use by another function.
	229	The Builder was unable to store to the DETSTAT CDS on the Builder point.
	230	The Builder was unable to \$synchronize delay while looping waiting for the Builder point to be built.
	231	The Builder was unable to initialize critical CDS to certain values due to Data Access errors.
	232	The requested checkpoint was not done because the Builder was active or on display at a User Station.
	233	A request for synchronization failed during a retry of a Data Access error on a Builder attempt to access the User Station node number in the Builder CDS.

Continued on next page

A.2 CLM Platform Error Codes, Continued

200-299 – Builder, continued

234	The Builder was unable to read the CDS parameter that contains the User Station node number that the Builder is displayed on. Int(2) is the SE_Status_enm value, Int(3) is the Minor_status_enm value, and Int(4) is the PA_status_enm value. Recall the Builder display.
235	The Builder was unable to store template names to the Builder CDS due to a Data Access error. Int(2) is the Minor_status_enm value and Int(3) is the entry number.
236	The Builder was unable to store template names to the Builder CDS due to a Data Access error. Int(2) is the Minor_status_enm_value and Int.(3) is the entry number.
237	The builder can not get memory to store the builder pointer for redundancy.

300-399 – Operator display task

300	Display Task unable to get memory for a type_value_rec after 1 minute of retries.
302	Error getting an NDT Entry
303	Call to Get_NDT_Entry returned wrong status
304	Failed attempting to restart a CLMI board
306	Error attempting to change state of a CLMI board
307-310	Bad Location Record passed to Find_NDT_Entry Int(3) has the page index and Int(4) has the entry offset from the location record
312	Error attempting to checkpoint network memory Int(3) is the status returned from CLM\$SAVE_NETMEM_CHKPT
313	Error on automatic checkpoint of network memory Int(3) is the status returned from CLM\$SAVE_NETMEM_CHKPT Int(4) is the current number of minutes past midnight
314	Error uploading/downloading from/to a port on call to Send_Msg Int(3) is the number of the port (CLMI #1 1-5; CLMI #2 10-14)
315	Error uploading/downloading from/to a port on call to Find_Cl_Req Int(3) is the number of the port (CLMI #1 1-5; CLMI #2 10-14)
321	Failed attempting to get File Manager Request Block
322-324	Call to get the address of a network memory entry returned a NIL pointer
325	Failed attempting to open a file Int(3) is FM_Status
326	Failed in attempt to change state of an NDT port, device, etc.
327	Failed in attempt to acknowledge the failure of a port, device, etc.
328-330	Call to network memory routine to get an entry location failed
331-335	NIL pointer found in active_display table info entry
336-339	Call to network memory routine to get an entry location failed Int(3) has the number of the display record being processed
341	Failed attempting to get File Manager Request Block
342	Invalid format in a record in a Device Status Definition File (.ZS) Int(3) is field number. Record Number not shown
343	Invalid format in a record in a Device Status Definition File (.ZS). A parenthesis "(" was expected Int(3) is field number. Record Number not shown
344	Invalid format in a record in a Device Status Definition File (.ZS). A quotation mark (") was expected Int(3) is field number. Record Number not shown

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A.2 CLM Platform Error Codes, Continued

300-399 – Operator display task, continued	347	Error Reading Device Status Definition File Int(3) is record number Int(4) is FM_Status
	348	Error while converting a 16-bit integer to ASCII string
	350	Error Fetching the STACDE parameters while checking the status of User Stations Int(3) is \$d_minor_status Int(4) is \$d_pa_status from the type value record
	351	Convert Entity Id Error Int(3) is \$d_se_status Int(4) is \$d_minor_status
	352	Error Converting CDS parameter names from External to Internal Int(3) is \$d_se_status Int(4) is \$d_minor_status
	353	Error Fetching a single number parameter Int(3) is \$d_minor_status Int(4) is \$d_pa_status from the type value record
	354	Error Fetching a single string parameter Int(3) is \$d_minor_status Int(4) is \$d_pa_status from the type value record
	355	Error Storing a single number parameter Int(3) is \$d_minor_status Int(4) is \$d_pa_status from the type value record
	356	Error Storing a single logical parameter Int(3) is \$d_minor_status Int(4) is \$d_pa_status from the type value record
	357	Error Creating IDB Int(4) is \$d_minor_status
	358	Error Adding item to IDB Int(4) is \$d_minor_status
	359	Error Collecting an IDB Int(3) is \$d_se_status Int(4) is \$d_minor_status
	360	Error Storing an IDB Int(3) is \$d_se_status Int(4) is \$d_minor_status
	361	Error Deleting an IDB Int(4) is \$d_minor_status
	362	Error doing Put_Memory of an Add List or a Type Value Record
	363	Error initializing the Add List Record
	364	Error trying to set NBUFER CDS Parameters to zero Int(3) is \$d_minor_status Int(4) is \$d_pa_status from the type value record
	365	Error trying to set STACDE CDS Parameters to zero Int(3) is \$d_minor_status Int(4) is \$d_pa_status from the type value record
	366	Error Storing a single string parameter Int(3) is \$d_minor_status Int(4) is \$d_pa_status from the type value record

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A.2 CLM Platform Error Codes, Continued

300-399 – Operator display task, continued

367	Error Fetching a single logical parameter Int(3) is \$d_minor_status Int(4) is \$d_pa_status from the type value record
368	Error Fetching a single enumeration parameter Int(3) is \$d_minor_status Int(4) is \$d_pa_status from the type value record
369	Error Storing a single enumeration parameter Int(3) is \$d_minor_status Int(4) is \$d_pa_status from the type value record
370	Error collecting an IDB Int(2) is the index of first add_rec item with an error. Int(3) is the pa_status from the first add_rec item to indicate an error.
371	Error storing an IDB Int(2) is index of first add_rec item with an error Int(3, 4) contain the decimal encoded address of the add_rec.
369	Error Storing a single Self Defining enumeration parameter. Int(3) is \$d_minor_status. Int(4) is \$d_pa_status from the type value record.

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A.2 CLM Platform Error Codes, Continued

400-499 – Message processor

400	Init_Queue failed due to problem in Get_Memory_Page
401	Deque detected a message that went past the tail
402	Deque received a strange response back from B\$Get_Memory_From_Pool
405	Find_CL_Req found an old message in the CL_REQ_TABLE
406	Find_FILE_Req found an old message in the FILE_REQ_TABLE
407	Find_PIT_Req found an old message in the PIT_REQ_TABLE
408	Find_PIT_Req found an old message in the PIT_REQ_TABLE and the message has not been sent to the CLMI board.
410	Msg_Hndlr hit the “otherwise” clause on MSG_TYPE
411	Msg_Hndlr received an invalid message from Deque
412	Deque returned an unexpected code to Msg_Hndlr
420	Send_Msg was unable to find PIT_Req in PIT_REQ_TABLE
421	Send_Msg was unable to find FILE_Req in FILE_REQ_TABLE
422	Send_Msg_Hndlr got an error return from Enqueue ** The message was discarded!
423	(Clmi_Req_Hndlr) An invalid NDT location was specified in CLMI message. Int(3) is the ndt page number. Int(4) is the word offset to the NDT.
430	PIT_Req_Hndlr was unable to find PIT_Req in PIT_REQ_TABLE
440	CL_Req_Hndlr hit the “otherwise” clause on REQ_TYPE
441	CL_Req_Hndlr encountered some error in processing the response
442	CL_Req_Hndlr (Move_Resp_To_Buffer) received a strange response back from B\$Get_Memory_From_Pool
450	Return_Error_To_CLMI was unable to send an error message back to the CLMI Int(2) is the return status from Send_Msg. Int(3) is the message type (i.e., 0=PIT, 1=CL, 2=CLMI, 3=FILE). Int(4) is the value in the first data word (16-bits) on the message.
451	CLMI_Req_Hndlr (Add_Entry_To_Table) hit the “otherwise” clause on REQ_TYPE
452	CLMI_Req_Hndlr encountered some error in processing the response
453	CLMI_Req_Hndlr could not do the requested status change ** The message was discarded!
454	CLMI_Req_Hndlr could not do the requested state change (not a FAILED state) ** The message was discarded!
455	CLMI_Req_Hndlr could not do the requested state change to FAILED state ** The message was discarded!
456	CLMI_Req_Hndlr hit the “otherwise” clause on REQ_TYPE
457	CLMI_Req_Hndlr was unable to get memory for rebuild_ALUT message
458	CLMI_Req_Hndlr was unable to send rebuild_ALUT message to Netmem_Handler
459	CLMI_Req_Hndlr received a message from a CLMI with a return status other than NO_ERROR(0).
460	FILE_Req_Hndlr (Get_IO_Number) could not find an available \$IO_Number
461	Msg_Hndlr (Complete_File_Req) hit the “otherwise” clause on req_type

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A.2 CLM Platform Error Codes, Continued

400-499 – Message processor, continued

462	Send_Msg returned a failure to Msg_Hndlr (Complete_File_Req)
463	FILE_Req_Hndlr hit the “otherwise” clause on O_OPEN_OPTION
464	FILE_Req_Hndlr hit the “otherwise” clause on REQ_TYPE
465	CL_MSG_BUFF_NOT_AVAILABLE can not reserve a buffer for CL message
466	Res_Buffer unable to Get_Memory for CL buffer
467	Res_Buffer unable to Get_Memory for CL buffer table
468	EMPTY_BUFF_ERROR in CLM\$Put_Field
469	INVALID_BUFFER_NUM in CLM\$Put_Field
470	INVALID_BUFFER_NUM in CLM\$Send_CL_Message
471	EMPTY_BUFF_ERROR in CLM\$Send_CL_Message
472	MISMATCH_BUFFER_SIZE_ERROR in CLM\$Send_CL_Message
473	INVALID_MSG_ID in CLM\$Recv_CL_Message
474	MSG_ID_NOT_FOUND in CLM\$Recv_CL_Message
475	MSG WAS DELETED, AFTER ONE HOUR TIME OUT
476	MSG STATUS STILL IN PROGRESS, AFTER ONE HOUR
477	ERROR OCCURRED DURING THE WAIT FOR MSG_STA
478	GLOBAL POINTER ERROR IN Send_CL_Message
479	Message time out in \$Recv_CL_Message.
480	Invalid message in CLM\$Recv_CL_Message.
481	Error Performing a File Operation for a CLMI Application. Int(2) is the type of operation: 0=open, 1=close, 2=delete, 3=read, 4=write, 5=rename. Int(3) is a sub-code. Int(4) is the RNOS File Manager Status.
482	No Memory in Heap Pool 2 available for store_pit request from the CLMI board. Int(3) is the status from B\$get_memory. Int(4) is the amount of memory requested.
483	The Message Processor attempted to move a request from the PIT Request Table that was of invalid request type. Int(2) is the index of the entry in the PIT Request Table. Int(3) is the invalid request type. Int(4) is the Message Id in the PIT Request entry.

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A.2 CLM Platform Error Codes, Continued

500-549 – Scheduler	500	Scheduler was unable to find page 1 of NETMEM
	501	Scheduler was unable to establish \$Periodic
	502	Scheduler was unable to get start of PIT directory page
	503	An invalid PIT/PITG location was specified
	504	Scheduler was unable to get start of PIT directory page (for loopback)
	505	An invalid PIT location was specified in a PITG
	506	Scheduler looped through PIT more than one time
	507	Scheduler could not get memory for build_complete pit for non-in-service device (aka FAIL_PIT)
	510	CLMI_Checker could not find the NDT Entry for a specific CLMI board
	511	CLMI_Checker could not FAIL a CLMI board
	512	AM is unable to communicate with CLMI board. Int(2) is the CLMI Board Number (i.e., 1 or 2)
	513	AM to CLMI Send Queue has been full for an excessive time. Int(2) is the CLMI board number (1 or 2).
	515	Change Priority of a task failed. Int(2) is the newly requested priority. Int(3) is the current priority or a zero. Int(4) indicates the instance of the change priority request.
	516	Scheduler could not get memory for priority list record.
	517	A bad pointer to an NDT entry was acquired. Int(3) is the NDT page number. Int(4) is the word offset to the entry on the NDT page.
	518	Unable to get the scheduler cycle parameter. INT(4) is the minor status enumeration ordinal error code.
	519	Unable to get the scheduler cycle parameter. INT(4) is the minor status enumeration ordinal error code.
	520	Unable to convert the CLM_nn entity id to fetch the scheduler cycle parameters. Int(4) is the minor status enumeration ordinal error code.
	521	Unable to get memory for converting schedule cycle params
	522	Unable to get memory for converting schedule cycle params
	523	Unable to convert STBUFFER parameter to internal. Int(3) is minor status ordinal. INT(4) in EXT/INT error status ordinal.

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A.2 CLM Platform Error Codes, Continued

550-590 – Synch processor

551	Send_Message_To_Synch_Proc failed trying to send message (\$Produce).
552	An invalid message was consumed—either wrong size or wrong type
553	Wait_For_Message failed trying receive a message (\$Consume)
555	Synch_Processor could not Get_Memory_Page (NDT_LUT_PAGE)
561	Synch_Processor could not get memory for Synch_Proc_Msg_Rec
562	An invalid NDT location was found on the requested device to synch
563	An invalid NDT location was found while attempting synch
564	Synch_Processor was unable to get start of PIT directory page
565	An invalid PIT/PITG location was specified
566	Synch_Processor could not get memory from Heap Pool 2 for Synchronization request. Int(3) is the status fro B\$get_memory. Int(4) is the amount of memory requested.
567	An invalid NDT location was found on the requested device to fail or place out-of-service
568	An invalid NDT location was found while attempting to make device status unknown
569	Synch_Processor hit the “otherwise” clause on state_change
570	An invalid NDT location was found on the requested to change states
571	Synch_Processor hit the “otherwise” clause on status_change
572	An invalid NDT location was found on the requested device to complete synch
573	An invalid NDT location was found while attempting synch completion
574	Synch_Processor could not determine the beginning of PIT directory page
575	An invalid PIT/PITG location was specified
576	Synch_Processor (Synch_Device) was unable to find known NDT location for CLMI in look-up table
577	Synch_Processor (Synch_Device) was unable to find known NDT location for Port in look-up table
578	Synch_Processor (Get_CLMI_And_Port) was unable to find entry for Port associated with device
579	Synch_Processor (Get_CLMI_And_Port) was unable to find entry for CLMI associated with device
580	Synch_Processor (Chg_State) was unable to find NDT location for primary of known device
581	Synch_Processor (SP_Change_State) was unable to find NDT location for primary of known device
582	Synch_Processor (Synch_Initialize) was unable to find NDT location for primary of known device
590	Change of NDT Status to Failed has occurred Int(2) is task id of the custom task that requested the failure. The task ids for various tasks are listed at the end of this list Int(3) is the NDT page number; Int(4) is the word offset into the NDT page

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A.2 CLM Platform Error Codes, Continued

591 - 599 – Redundancy CLM

591	CLM failover due to a failure of an NDT (which was configured for failover).
592	CLM can not fail over b/c the secondary node is not alive.
593	CLM failover has put memory error.
594	CLM failover has a DUPLICATE_POINTER_NAME error.
595	CLM failover has a GLOBAL_MEMORY_TABLE_FULL error.
596	CLM failover can not locate netmem pointer.
597	CLM FAIL, can not clear network memory, in failover.

600-639 – Store intercepts

600	Intercept error PRODUCE_ERROR in produce_message
601	Get memory error for PPS parameter in store_intercept_stub OPERATOR STORE
602	Get memory error for CLM_STORE parameter in store_intercept_stub
605	Get memory error for CLM_OUT parameter in store_intercept_stub
606	Get memory error for CLM_PPS parameter in store_intercept_stub
607	Get memory error for CLM_OUT parameter in fast_cds_stub
608	Get memory error for CLM_PPS parameter in fast_cds_stub
611	Error CONSUME_ERROR in consume message
612	An invalid NDT location was specified in the PITE for which a store is being attempted.

700-799 – Network memory interface routines

700	Get_Memory_Page: Detected circular reference on page
701	Initialize_Network_Memory: Network memory was already initialized
702	Initialize_Network_Memory: Could not allocate the memory necessary for the page index table
703	Initialize_Network_Memory: Could not allocate the memory necessary for the reserved memory pages
704	Allocate_Memory_Page: Network memory not initialized
705	Allocate_Memory_Page: Page table full error
706	Allocate_Memory_Page: Given page index is already allocated
707	Allocate_Memory_Page: Unable to allocate memory for new page
708	Deallocate_Memory_Page: Network memory not initialized
709	Deallocate_Memory_Page: Unable to produce message to netmem_handler
710	Purge_Deallocated_Pages: Network memory not initialized
711	Purge_Deallocated_Pages: Put memory failed
712	Get_Memory_Page: Network memory not initialized
713	Get_Memory_Page: Given page index is invalid
714	Switch_Page: Network memory not initialized
715	Switch_Page: One or Both of the given pages is invalid
716	Clear_Network_Memory: Clearing all network memory. FOR TEST ONLY
717	Clear_Network_Memory: Network memory not initialized
718	Clear_Network_Memory: Put memory failed
719	Allocate_Memory_Page: Given page index is invalid
720	Add_Entry: Entry size is invalid
721	Check_Location_Rec: Location offset is invalid
722	Check_Location_Rec: Entry size is invalid
731	Netmem_Handler: Job deleted during consume timeout

Continued on next page

A.2 CLM Platform Error Codes, Continued

700-799 – Network memory interface routines

732	Purge_Deallocated_Pages routine was unable to get the checkpoint semaphore. Int(2) is the return status from \$get_semaphore.
733	Purge_Deallocated_Pages routine was unable to release the checkpoint semaphore. Int(2) is the return status from \$release_semaphore.
734	Rebuild_PPA_LUT routine was unable to get the checkpoint semaphore. Int(2) is the return status from \$get_semaphore.
735	Rebuild_PPA_LUT routine was unable to release the checkpoint semaphore. Int(2) is the return status from \$release_semaphore.
736	Rebuild_PITE_LUT routine was unable to get the checkpoint semaphore. Int(2) is the return status from \$get_semaphore.
737	Rebuild_PITE_LUT routine was unable to release the checkpoint semaphore. Int(2) is the return status from \$release_semaphore.

800-899 – Miscellaneous routines

800	\$D_Send_Event encountered a problem
801	Report_Error could not convert CLM point entity name (to compute unit_number)
802	Report_Error was unable to send a Status Notification journal message due to a bad return status from the Status Accountant. INT(3) contains the Status Accountant error code.

900-999 – PIT processor (fetcher/storer)

900	Fetcher was unable to get memory for FETCH_TABLE
901	Fetcher was unable to get memory for an IDB
902	Fetcher was unable to establish \$Periodic
903	Fetcher encountered error in \$Cycle
904	Fetcher encountered an invalid message type ** The request was discarded!
910	(Check_Fetch_Param) An invalid PITE Parameter location was specified
911	Fetcher encountered a problem in \$D_Create_IDB
912	Fetcher encountered a problem in \$D_Add_IDB_Items
913	Fetcher encountered a problem in B\$Store_Collect_IDB
914	Fetcher encountered a problem in \$D_Open_Idb
915	Fetcher encountered a problem in \$D_Close_Idb
916	Fetcher (Send_Completed_Requests) hit the “otherwise” clause on msg_type
917	Storer could not clear synch due to an invalid PIT address
920	(Calculate_Port) An invalid NDT location was specified
921	(Check_NDT_STS_STA) An invalid NDT location was specified
924	An invalid PITG location was specified
925	An invalid PIT location was specified
926	PIT_Processor could not find First PITE in PIT
927	PIT_Processor could not find Next PITE In PIT
928	An invalid PITE location was specified
929	PIT_Processor could not find PITE Parameter in PITE
941	Storer was unable to get memory for External point names

Continued on next page

A.2 CLM Platform Error Codes, Continued

900-999 – PIT processor (fetcher/storer), continued

942	Storer was unable to get memory for Internal point names
943	Storer was unable to convert External to Internal point names
950	Storer was unable to get memory for store_table
951	Storer was unable to get memory for an IDB
952	Storer was unable to establish \$Periodic
953	Storer encountered error in \$Cycle
954	Storer encountered an invalid message type ** The request was discarded!
960	(Check_Store_Param) An invalid PITE Parameter location was specified
961	Storer encountered a problem in \$D_Create_IDB
962	Storer encountered a problem in \$D_Add_IDB_Items
963	Storer encountered a problem in B\$Store_Collect_IDB
964	Storer encountered a problem in \$D_Open_Idb
965	Storer encountered a problem in \$D_Close_Idb
966	Storer (Send_Completed_Requests) hit the “otherwise” clause on msg_type
967	Storer could not complete synch due to an invalid PIT address
968	Storer could not clear synch due to an invalid PIT address
969	(Check_Store_Param) A PITE parameter input TYPE_VALUE_PTR could not be found
970	Storer detected a PIT request over 2 hours old and discarded it. Int(2) is the PIT Request Table entry number. Int(3) is the ID of the message sent to the CLMI board. Int(4) is the current status of the request.
971	Storer detected a PIT request over 2 years old and was never sent to the CLMI board. Int(2) is zero. Int(3) is the PIT Request Table entry number. Int(4) is the current status of the request.
999	A CLM Platform task was unable to get AM memory for a particular function.

–9999-0

Interface board reported errors

A.3 CLM Operational Messages

During normal operation, certain messages may be journalized to the Operator Message journal or on R400 and later systems, the Status Notification journal.

The following messages may be generated by the CLM operating system. The message is followed by a short description.

CURRENT BUILDER ACTION ABORTED DUE TO UNEXPECTED EXIT FROM DISPLAY.

The user has left the Configuration Builder display while a Builder process was still executing. The Builder process has been aborted.

BUILDER POINT DOES NOT EXIST. BUILDER TASK STARTUP INCOMPLETE.

The CLM has started up, but has not detected the required Builder image point BUILD_nn, where nn = CLM node id. The Builder task is not yet operational.

BUILDER POINT NOW EXISTS. BUILDER TASK STARTUP IS COMPLETE.

The Builder task has detected the Builder image point and now completed the task startup. The Configuration Builder may now be used.

RECV_CL_MESSAGE FAILED, THE RETURN STATUS IS NNNN

The CL routine has encountered an error during processing and has returned an error code nnnn. Reference the *CLM Platform Implementation Manual*.

CLM\$GET_FIELD FAILED, THE RETURN STATUS IS NNNN

The CL routine has encountered an error during processing and has returned an error code nnnn. Reference the *CLM Platform Implementation Manual*.

RESERVE BUFFER FOR SEND OR RECV MSG FAILED, THE RETURN STATUS IS NNNN

The CL routine has encountered an error during processing and has returned an error code nnnn. Reference the *CLM Platform Implementation Manual*.

RELEASE BUFFER FOR SEND OR RECV MSG FAILED, THE RETURN STATUS IS NNNN

The CL routine has encountered an error during processing and has returned an error code nnnn. Reference the *CLM Platform Implementation Manual*.

Continued on next page

A.3 CLM Operational Messages, Continued

CLM\$PUT_FIELD FOR SEND OR RECV MSG FAILED, THE
RETURN STATUS IS NNNN

The CL routine has encountered an error during processing and has returned an error code nnnn. *Reference the CLM Platform Implementation Manual.*

CLM\$SEND _CL_MESSAGE FAILED, THE RETURN STATUS IS
NNNN

The CL routine has encountered an error during processing and has returned an error code nnnn. *Reference the CLM Platform Implementation Manual.*

FIELD INPUT CHANGED: PPPPPPPP NEW VALUE: VVVVV.VV

A field input value for parameter PPPPPPPP has changed. The new value is VVVVV.VV

PPPPPPPP IS BEHIND.

The PIT named PPPPPPPP is behind in its schedule processing. It may be necessary to check the pit loading.

DDDDDDDD STATE CHANGE TO IN_SERV

The device DDDDDDDD has changed its state to IN SERVICE and is now fully communicating.

DDDDDDDD STATE CHANGE TO SYNC

The device DDDDDDDD has changed its state to SYNC and is currently synchronizing data between the device and the CLM Network Memory.

DDDDDDDD STATE CHANGE TO FAILED

The device DDDDDDDD has failed.

DDDDDDDD STATE CHANGE TO OUT_SERV

The device DDDDDDDD has been placed out of service and is not processed.

DDDDDDDD STATUS CHANGE TO OK

The status of the device DDDDDDDD is now OK and fully communicating.

DDDDDDDD STATUS CHANGE TO ERROR

The status of the device DDDDDDDD has changed to ERROR and has missed at least one poll cycle.

DDDDDDDD STATUS CHANGE TO FAILED

The status of the device DDDDDDDD has changed to FAILED indicating that communication is not currently possible.

Continued on next page

A.3 CLM Operational Messages, Continued

DDDDDDDD STATUS CHANGE TO ACK_FAIL

The operator has acknowledged the failed device DDDDDDDD from the CLM displays.

DDDDDDDD STATUS CHANGE TO UNKNOWN

The device DDDDDDDD has changed its status to Unknown because the device state is Out of Service.

DDDDDDDD STATUS CHANGE TO PRI_UNAV

The device DDDDDDDD has changed its status due to a communication error with a higher level device.

BOARD nn FOR NDT DDDDDDDD RETURNED TO NORMAL

Device DDDDDDDD has had a previously failed board slot nn return from an error condition.

BOARD nn FOR NDT DDDDDDDD INVALID SUBDEV STS

Device DDDDDDDD has had a subdevice status change for board slot nn.

QUESTIONABLE SYNCHRONIZATION OCCURRED ON PIT

PPPPPPPP - RESYNCH SUGGESTED

The CLM has detected a questionable synchronization between the PIT PPPPPPPP and its associated device. It is suggested that the device be placed Out of Service and then back In Service.

PARAMETERS MAY NOT HAVE SUCCESSFULLY SET TO NaN FOR PIT

PPPPPPPP

The CLM has detected that certain parameters in the associated PITEs may not have been set BAD when the PIT was unable to collect data.

LOAD FAILED ATTEMPTING TO SEND INITIAL LOAD PACKET TO CLMI BOARD

The CLM has detected an error while attempting to download the CLMI board operating system.

LOAD FAILED, SYSTEM ERROR 116 HAS RECORD NUMBER (3) AND LOADER STATUS (4).

The CLM has detected an error while attempting to download the CLMI board operating system. Reference the System Error journal for detailed error data.

Continued on next page

A.3 CLM Operational Messages, Continued

UNEXPECTED END OF FILE IN THE OBJECT FILE, OBJECT FILE
ERROR!

The CLMI Loader function has detected an unexpected end of file while attempting to download the application file to the CLMI board. Possible corrupted file.

LOAD FAILED DUE TO INVALID "CLMI" DIRECTIVE COMMAND.

An error in the executable statements in the CLMI configuration file has caused the download to the CLMI board to abort. Check the .ZC file on the HM.

THE REQUESTED OBJECT FILE FFFFFFFF.S IS NOT FOUND!

The CLMI loaded function has not been able to find the application file FFFFFFFF.S. Check the HM and the &CUS directory.

LOAD FAILED DUE TO INVALID "LOAD" DIRECTIVE COMMAND.

An error in the executable statements in the CLMI configuration file has caused the download to the CLMI board to abort. Check the .ZC file on the HM.

LOAD FAILED DUE TO INVALID "INIT" DIRECTIVE COMMAND.

An error in the executable statements in the CLMI configuration file has caused the download to the CLMI board to abort. Check the .ZC file on the HM.

LOAD FAILED DUE TO INVALID "RUN" DIRECTIVE COMMAND.

An error in the executable statements in the CLMI configuration file has caused the download to the CLMI board to abort. Check the .ZC file on the HM.

LOAD FAILED DUE TO INVALID "END" DIRECTIVE COMMAND.

An error in the executable statements in the CLMI configuration file has caused the download to the CLMI board to abort. Check the .ZC file on the HM.

LOAD FAILED DUE TO INVALID CONFIGURATION FILE.

An error in the executable statements in the CLMI configuration file has caused the download to the CLMI board to abort. Check the .ZC file on the HM.

THE REQUESTED CONFIGURATION FILE CLMI_{nnn}.ZC IS NOT
FOUND !!!

The CLMI Configuration file has not been found for the specified CLM on the HM in the &CUS directory. FORM : CLMI_{nnn}.ZC, where nnn = CLM node id.

CHECK POINT IS NOT SAVED! NOT ENOUGH MEMORY OR
ERROR OCCURRED!

The CLM checkpoint function has not completed due to an encountered error. Reference the System Error journal for details.

A.4 Messages from Recoverable Errors (BSAP)

0-999

The following messages come from the Bristol-Babcock BSAP application program on the Interface board and are recoverable. These do not cause a board crash.

21	Internal Error 21. No RRH Queue Slots
143	Internal Error 143. Invalid request from AM
243	Internal Error 243, get_pitep_ista failed
245	PITE Conversion Code Invalid
262	Internal Error 262, alarm input string from RTU not null terminated
265	Internal Error 265, could not point to alarm data in alarm message buffer
300	Poll Task: No NDT for Device Address:
301	Poll Task: Key Number Out of Range:
302	Poll Task: Destination Size Error:
303	PC Tasks: No NDT for Device Address:
304	PC Tasks: Key Number Out of Range:
305	PC Tasks: Destination Size Error:
306	Master Tasks: No NDT for Device Address:
307	Master Tasks: Key Number Out of Range:
308	Master Tasks: Destination Size Error:
309	Comm Drivers: No NDT for Device Address:
310	Comm Drivers: Key Number Out of Range:
311	Comm Drivers: Destination Size Error:
312	Comm Buffer Pool Error:
313	Download complete to device at address:
413	Internal Error 413; Device Data Response in Multiple Messages
418	Internal Error 418; Conversion Code Parse Failed
422	Internal Error 422; Version Number Error
423	Internal Error 423; Detailed Status Storage Failed
425	Internal Error 425; NDT Address Retrieval Failed
426	Internal Error 426; Version Number Storage Failed
429	Internal Error 429; MSD Storage Failed
430	Internal Error 430; Conversion Code Retrieval Failed
433	Internal Error 433; Mismatch with Count of Parameters and Elements
437	Internal Error 437; Convert and Store Failed
439	Internal Error 439; Data Response Muddle
440	Internal Error 440; Not Enough Data in Response to Satisfy Parameters
441	Internal Error 441; Data Left in Response after Parameters Satisfied
500	Internal Error 500; RTU address greater than maximum configured RTU
999	Internal Error

* Nonrecoverable

Continued on next page

A.5 Codes from Nonrecoverable CLMI Error (BSAP)

Message format

The Interface board must be restarted following any nonrecoverable error. Each nonrecoverable event is added to the TPN operator message journal in the following format:

CLMI: Bristol Interface Failed: Int(1) Error, Int(2) Task, Int(3) PC

where:

Int(1) Integer(s) equal to the error code listed below

Int(2) Integer(s) equal to the CLMI task number

Int(3) The Program Counter

Utility task

nonrecoverable errors

1	Mail connect failure
2	Cannot get heap
3	Router connect status equals failure
4	Configuration failure RTU Ports not contiguous
5	No PC port specified
6	To few RTU ports specified
7	To many PC ports defined
8	Mail send error on pseudo connect
9	Pseudo mbox id failure
10	Did not find pseudo port
11	No PC task start
12	No Request Return Handler Start
13	No Sync Processor start
14	No CLM Req Handler Start
15	No poll task start
16	No Master task start
17	Tick-tock event flag error
18	Release connect msg memory error
19	Unable to send File open mail
20	NRT write fail
21	No RRH queue slots available
22	No Alarm Processor start

CLM Request Handler task nonrecoverable errors

100	Read of input mailbox fail
101	Send of illegal req to CLM fail
102	Send retry message fail
103	CLM req handler no heap
104	Sync mailbox fail

Continued on next page

A.5 Codes from Nonrecoverable CLMI Error (BSAP), Continued

Alarm handler task nonrecoverable errors	207	Send fatal error
	211	Pause fatal error
	240	Could not get key parameter for RTU
	241	Could not get state for RTU
	261	Receiving of Bristol alarm message failure
Request return handler task nonrecoverable errors	401	Free memory fatal error
	402	Set event flag delay fatal error
	403	Clear event flag fatal error
	404	Wait event flag fatal error
	405	Get semaphore fatal error
	406	Release semaphore fatal error
	407	Send mail fatal error
	408	Receive mail fatal error
	409	Pause task fatal error
	410	Outstanding request block counter error
	411	Invalid rrh_list location
	412	Request ALUT type in list type missing
	414	Invalid list type in header block
	415	Invalid request type in request block
	416	Expected AM message missing
	417	Unexpected pointer to AM message
	419	Unexpected pointer to first request block
	420	Request block's pointer to header block is NULL
	421	No msg_id match for ALUT complete message
	424	Retrieval of next PITEP failed
	427	Retrieval of PIB offsets failed
	428	Retrieval of next PITE failed
	431	Retrieval of NDT key parameter failed
	432	Unknown RTU code
	434	Number of requests returned exceeds number of requests sent
	435	Bad PIB in request block
	436	Received invalid mail message
	438	Setreq for RTU response failed
	442	Invalid request block pointer
	443	Invalid outstanding request block counter array index
	444	Can't determine if RTU is configured for RBE

Continued on next page

A.5 Codes from Nonrecoverable CLMI Error (BSAP), Continued

**Report by exception
task nonrecoverable
errors**

600	Unable to retrieve BSAP packet
601	Unable to clear RBE event flag
602	Error on waiting for event flag
603	Error reading mailbox
604	Undefined CLM message type
605*	Error getting scan period parameter
606	Error setting event flag delay
607	Error getting state key
608	Error putting state key
609	Error getting NDT
610*	Communication error in RBE message
611*	Received unknown message type
612*	RBE configuration error (RTU was configured to GO ACTIVE)
613	Unable to save report message sequence number in NDT
614	Error accessing detail status
615	Error accessing node state
616	Error getting poll rate
617*	Error getting NDT key

* Recoverable error.

**Error Message
Handler**

999	Undefined error message
-----	-------------------------

A.6 Device Error Codes on PITE Display (BSAP)

The following table lists error codes that may show up on the CLM PITE Detail display, Fig. 1-14, These codes will be listed under the ERROR CODE columns.

Codes listed

	No error shows up as blank
1	Error - nonspecific (Workstation Display)
3	Unknown

1001	Data response in Multiple Packets
1002	BSAP error: attempt to write to read only area
1004	BSAP error: invalid identifier, i.e, name, list or invalid MSD address
1005	BSAP error: security violation
1006	BSAP error: attempt to write to manual inhibited signal
1007	BSAP error: invalid write attempt, i.e, set analog signal ON
1010	BSAP error: no match on signal name

1104	BSAP error: no match found on name
1108	BSAP error: analog value found in packed logical expression
1116	BSAP error: no match found
1132	BSAP error: version number mismatch
1164	BSAP error: invalid database function requested

1200	Questionable bit set in analog value
1201	Unable to obtain conversion string
1202	Unable to parse conversion code
1203	Data conversion and storage failed
1204	No data left to store for this parameter
1205	Disagreement between BSAP/LCN data types
1206	Parsing of BSAP data hopelessly muddled
1207	Invalid BSAP type code or data muddle
1208	No MSD found for parameter data request
1209	Send request failed
1210	No version number or version mismatch on ACCOL load
1211	Timed out waiting for BSAP response
1212	Get NDT address failed
1213	Problem getting NDT address
1214	Get device status failed
1215	Version number mismatch on this parameter
1216	Parameters requested do not match BSAP data
1217	Problem parsing BSAP message
1218	Unreadable BSAP response
1219	Error during synchronization

Continued on next page

A.7 Error Codes on Check Before Operate (CBO) (BSAP)

Codes listed

1220	Problem parsing BSAP header
1221	Device is unavailable
1222	Timed out waiting for parameter address from RTU
1223	Timed out waiting for data during synchronization
1224	Communication failure when attempting to read out request

Codes 2000 - 2034 are reserved for CBO return error codes.

0 - CBO good status.

2000 - Base offset for CBO return error codes

2003 - Device Error, board missing or incorrect type.

2008 - Select/Check read back error - select failed

2009 - Operate/Check error - operate failed.

2011 - Timeout or no select in progress. Unmatched transaction code.

2012 - Configuration error. No elements in request.

2013 - Multiple select request to CBO board. Previous select pending on requested board.

2014 - Power lost between select and operate requests.

2015 - CBO board error already exists.

2016 - Invalid manual CBO request function code.

2032 - No match of first MSD to any manual CBO module.

2033 - Additional MSD's not all on same CBO board.

2034 - Operate request does not match select request.

2035 - Trailing edge of pulse failed.

2201 - Parameter status for sndmsg error

2202 - PIB setup error.

2203 - GET MSD ADDRESS ERROR.

2204 - Error on get output conv code.

2205 - Error getting output value.

2206 - Error getting next pitep.

2207 - Incorrect RTU response.

2208 - CBO request timeout.

2209 - Too many CBO parameters on PITE.

2210 - Error getting board list.

2211 - Error getting board info.

2212 - Bad BCO board number or not found.

A.8 Nonrecoverable Error Codes Modbus

Nonrecoverable Modbus error codes

During a nonrecoverable Modbus error, the following message is journalled:

CLMI: Modbus Interface FAILED: Error XXX, Task YYY, PC = ZZZ

Nonrecoverable Modbus Errors

Error XXX	Meaning
2	No memory left on the Interface board: Contact TAC.
207	SEND_UNSUCCESSFUL: Sending of a mail message to a particular mailbox failed. Contact TAC.
211	PAUSE_UNSUCCESSFUL: Pausing of a particular task did not succeed. Contact TAC.
501	TOO_MANY_PORTS have been configured for this application. The probable cause could be the INIT directive in the CLMI_XX.ZT file.
502	START_TASK_FAILED: The operating system failed to start one or more of the application tasks. Contact TAC.
503	RECEIVE_FAILED: A mail message was not received in one of the operating system mailboxes. Contact TAC.
504	ILLOGICAL_CONDITION: An illogical condition occurred during system operation. It would be inappropriate for the application to continue any further. Contact TAC.
505	CONNECT_FAILURE: The application during startup failed to link to a required mailbox. Contact TAC.
506	BUFFER_CORRUPTION: A communication buffer has been corrupted during operation. Contact TAC.
509	REC_BUFF_CORRUPTION: A communication buffer has been corrupted while receiving an incoming message. Contact TAC.
514	GET_SEM_ERROR: The application was unable to get a required semaphore. Contact TAC.
515	REL_SEM_ERROR: The application was unable to release an acquired semaphore. Contact TAC.
604	INVALID_CONFIGURATION: The application has not been configured properly. Possible Problems: • A read of a Modbus address has been requested and the corresponding data type in the device NDT is 0 (FC01, FC02, FC03, or FC04). • The diagnostic cycle key parameter does not exist. Was the port NDT built using M_PORT.ZT?

Note: When calling about the problem, have the task numbers and program counter address available.

A.9 Detail Status Error Codes Modbus

Modbus parameter Detail Status Error codes

Interface board parameter detail status display error codes. These errors will show up on the CLM PITE Detail Display, Fig. 1-14, under the ERROR CODE columns.

Detail Status Error Codes

Value	Meaning
601	Invalid Conversion Code
602	Invalid Modbus Address
603	Input Error - Possible Data Type Mismatch Between Conversion Code and Type Value Record
604	INVALID_CONFIGURATION: The application has not been configured properly. Possible Problems: • A read of a Modbus address has been requested and the corresponding data type in the device NDT is 0 (FC01, FC02, FC03, or FC04). • The diagnostic cycle key parameter does not exist. Was the port NDT built using M_PORT.ZT?
2001	Protocol Error
2002	Timeout Error
2003	Protocol, Timeout Error
2004	Checksum Error
2005	Protocol, Checksum Error
2006	Timeout , Checksum Error
2008	Unit Char or Buffer Error
2009	Protocol, Unit Char or Buffer Error
2010	Timeout, Unit Char or Buffer Error
2012	Checksum, Unit Char or Buffer Error
3001	Illegal Function Code
3002	Illegal Data Address
3003	Illegal Data Value
3004	Slave Device Failure

Appendix B – Rebuilding the Network Memory Database

B.1 Network Memory Database Rebuild Procedure

Procedure

There may be occasion to warrant a rebuild of the Network Memory database. The procedure to rebuild Network Memory follows.

Procedure to Rebuild Network Memory

Step	User Action
1	PERFORM A BACKBUILD OF THE EXISTING NETWORK MEMORY From the CLM Configuration Builder, • Select the BACKBUILD target • Enter the name of the file to direct the data output (The default pathname will be NET>&CUS) • Select the BUILD target • Verify successful completion
2	SAVE THE CLM CHECKPOINT FILE From the Command Processor in the Engineering Personality • SP NET>&CUS • COPY CLM0nn.CP SAVE SAV0nn.CP (where nn = node number of CLM)
3	SETUP FOR A NULL DATABASE RESTART OF THE CLM From the CLM Configuration Page, • Select the NULL LOAD option in the Pathname Configuration section From the AM System Status Display, • Perform a SAVE DATA for the CLM node
4	SHUT DOWN AND RELOAD THE CLM NODE(S) If the CLM node is redundant, both Primary and Backup nodes must be shut down. At load completion time, the CLM Network Memory should be a null database. This may be verified from the CLM Status displays where there should be no devices shown or defined.
5	INSTALL NEW APPLICATION FILES, IF REQUIRED At this point, any new application files (*.S) or template files (*.ZT) must be copied to the appropriate media.

Continued on next page

B.1 Network Memory Database Rebuild Procedure, Continued

Table B-1 Procedure to Rebuild Network Memory, continued

Step	User Action
6	REBUILD THE NETWORK MEMORY FROM BACKBUILD FILE From the CLM Configuration Builder, • Select the FILE INPUT target • Enter the name of the Backbuild file created in Step 1 earlier • Select BUILD to re-install the Network Memory database • Verify successful completion
7	ESTABLISH COMMUNICATIONS WITH DEVICE NETWORK Place all required devices back IN SERVICE and verify data is being collected.
8	RECONFIGURE THE CLM SAVE DATA PATH From the CLM Configuration Page, • Select the SAVE DATA pathname target in the Pathname Configuration section • Enter the directory name where the CLM Network Memory database will be checkpointed
9	SAVE THE NEW CLM DATA BASE From the AM System Status Display, • Perform a SAVE DATA for the CLM node
10	VERIFY SUCCESSFUL COMPLETION OF CLM SAVE DATA From the CLM Status Display (CLMSCHEM), select the CLM node number and confirm the proper Time/Date stamp at the bottom of the screen for the new CLM Network Memory database.

Appendix C – Analyzing CLM Failures

C.1 CLM Node Failure

CLM Node Failure

In the unlikely event of a CLM physical node crash, certain critical pieces of information must be passed back to Honeywell TAC so that effective analysis of the crash may occur. Some of this information must be obtained prior to the restart of the CLM node.

CLM Node Failure Analysis Requirements

Step	Description
1	Prior to the restart of the CLM node, PRINT THE DETAILED ERROR DATA FOR THE FAILURE From the Engineering Main Menu, • Select the SMCC/MAINTENANCE target • Select the MODULE ERROR target • Select the DETAILED MODULE ERRORS target • Enter the failed CLM node number • Select the PRINT target (may require changing to a valid printer id)
2	PRINT THE CRASH DATA REGISTERS From the Engineering Main Menu, • Select the SMCC/MAINTENANCE target • Select the PROBE FAILED MODULE target • Enter the failed CLM node number and address 84020 • When the data is displayed, select the PRINT target
3	PRINT THE APPROPRIATE JOURNALS From the Event History Menu, for the last hour, • Print the Status Notification Journals (or Status Change) • Print the System Error Journals
4	PERFORM A DUMP OF THE FAILED CLM NODE From the AM System Status Display, • Select the CLM node number • Select the DUMP NODE target • Direct the output to a Removable Media destination The Dump of the node is complete when the status of the CLM node changes to the POWER ON state.

Continued on next page

C.1

CLM Node Failure, Continued

5	After the dump of the CLM node is complete (Status = POWER ON), RESTART THE CLM NODE
6	OBTAIN THE NUMBER OF SYSTEM-USED CUSTOM TASKS From Page 2 of the LCN NODES function for the CLM node, - Record the number of Background CL Tasks - Record the number of Concurrent Data Access Tasks
7	OBTAIN THE CLM PLATFORM RELEASE NUMBER From the CLM Status Display (CLMSCHEM), - Select the CLM node number - Record the CLM Platform release number from the lower left corner of the display
8	OBTAIN THE CLM APPLICATION RELEASE NUMBER From the CLM Status Display (CLMSCHEM), - Select the CLM node number - Select the first implemented port - Select the DETAIL STATUS target - Upon display invocation, select the PORT target - Select the DETAIL STATUS target - Upon display invocation, record the application release number from the upper right part of the screen. Repeat for any different applications on other ports.
9	OBTAIN THE CLM LOAD ADDRESSES - Invoke the DETAIL DISPLAY for the BUILD_<i>nn</i> point id, where <i>nn</i> = the CLM node number (or lower redundant node) - PAGE FWD until the custom parameter PROCNAM is found - Record the two address values in the string parameter
10	OBTAIN THE TPS RELEASE NUMBER From the Engineering Main Menu, - Select the SMCC/MAINTENANCE target - Select the REV/CONFIG STATUS target - Enter the CLM node number - Record the SOFTWARE RELEASE NUMBER at the bottom of the screen
11	SEND ALL OF THIS DATA TO HONEYWELL TAC Honeywell Technical Assistance Center Phone: (602) 313-5558 or 1-800-822-7673

Appendix D – CLM Troubleshooting

D.1 CLM Platform Debug

CLM Platform Problems

On occasion, things may not work as expected. In most cases, this is an error in configuration or an incomplete installation. Some of the most common problems reported and corrective checks to make are listed in this section.

CLM Platform - Corrective checks for common problems

Problem	Corrective Checks
CLM node crashes in Display Task after node startup	See Section A.1. Check for the existence of the CLM_nn and BUILD_nn points. If either is missing, refer to the <i>CLM Platform Implementation Manual</i> for installation. Check for the total number of tasks defined for the CLM. Add the CLM tasks (9) plus Background CL tasks plus Concurrent Data Access tasks (from NCF node data, page 2). This total must be less than the maximum allowed for the specific TPS release. If exceeded, reduce the number of CL background tasks (Concurrent DA must be at 4 and CLM is not adjustable).
The Builder function does not work (i.e goes BUSY on command and never returns)	See Section A.1. Check for the total number of tasks defined for the CLM. Add the CLM tasks (9) plus Background CL tasks plus Concurrent Data Access tasks (from NCF node data, page 2). This total must be less than the maximum allowed for the specific TPS release. If exceeded, reduce the number of CL background tasks (Concurrent DA must be at 4 and CLM is not adjustable). Check NCF, page 3 of the node data to confirm the CONV and CLM external load modules are listed. If not, add to the table and do an on-line modification to the NCF data for the CLM node. Reload the node.
Invoking the DETAIL STATUS target after Port selection, gives message "Collecting Data - Please Wait" forever.	See Section A.1. Check for the total number of tasks defined for the CLM. Add the CLM tasks (9) plus Background CL tasks plus Concurrent Data Access tasks (from NCF node data, page 2). This total must be less than the maximum allowed for the specific TPS release. If exceeded, reduce the number of CL background tasks (Concurrent DA must be at 4 and CLM is not adjustable). Check NCF, page 3 of the node data to confirm the CONV and CLM external load modules are listed. If not, add to the table and do an on-line modification to the NCF data for the CLM node. Reload the node.

D.2 CLM Interface Board Debug

CLM Board Problems On occasion, things may not work correctly when attempting to perform an action on/to the CLM interface board. The following lists several problems and their corrective checks.

CLM Interface Board - Corrective checks for common problems

Problem	Corrective Checks
Unable to place board In Service. Status/State goes to FAIL/FAIL.	Check the CLMI0nn.ZC file. Refer to the <i>CLM Platform Implementation Manual</i> . Verify correct format of configuration statements. Verify the existence of the application file (*.S) and the board operating system file (CCP.S or CLMI.S). These files should exist in the directory that is specified in the PATHNAME CONFIGURATION section of the CLM Configuration page.
Changing the baud rate through the Builder is not being reflected in the Device Detail Status Display..	Check CLMI0nn.ZC file to be sure that the INIT statement has a "1" for the port number.
Modbus & Modbus Plus: Placing device In Service cause the Status/State to go to UNKNOWN/SYNC forever.	Check CLMI0nn.ZC file to be sure that the INIT statement has a "1" for the port number of the device.
Modbus & Modbus Plus: Placing the device In Service cause the interface board to fail with a "604" error in the journal.	Check the device Builder template entries for all function codes (FCnn) being used. If this number is zero and a request is made for data using that function code, then the board will fail. This field must have a positive number if the function code is being used.

Appendix E – Windows NT Event Log Messages

When problems are detected by the TPS/CLM Platform, messages are logged describing the problem to the Windows NT Application Event Log. The Event Logs may be viewed using the Event Viewer located under the *Administrative Tools (Common)* menu. Following are messages logged by the TPS/CLM Platform:

Message ID	Description
10400	LCN Node <i>nn</i> : CLMI Channel 1 Started. Indicates that the CLMI board emulation software has started successfully. The CLM Personality may now be loaded from the TPN.
10430	LCN Node <i>nn</i> : Error Connecting to Mailbox (Named Pipe). <i>nn</i> . Indicates that the TPS/CLM has an internal communications problem. This is usually a temporary problem that occurs when the CLMI board is put In-Service.
10431	LCN Node <i>nn</i> : An Unrecognized message was received by the TPS/CLM Platform. This is not unusual and normally causes no problems.
10432	LCN Node <i>nn</i> : Queue for Sending Messages to TPN from CLM Interface is Full. This means the TPN resident portion of the CLM Platform is not removing messages as fast as the Windows NT resident portion of the CLM Platform is inserting them. This is usually a temporary problem.
10460	LCN Node <i>nn</i> : Error Creating Mailbox (Named Pipe); Error: <i>nnnnn</i> . This is an Internal TPS/CLM Platform error; contact Honeywell Technical Assistance.
10461	LCN Node <i>nn</i> : Could not Connect to CLM Message Router MailBox; Error: <i>nnnnn</i> . This is an Internal TPS/CLM Platform error; contact Honeywell Technical Assistance.
10462	LCN Node <i>nn</i> : Too Many Ports Configured; see KEY_LOCAL_MACHINE\Software\Honeywell\Tdc Emulators\TDCBoard0\CLM\<app>\INIT value in the Registry.
10463	LCN Node <i>nn</i> : Failed Reading from MailBox; Error: <i>nnnnn</i> . This is usually a temporary problem. If it persists contact Honeywell Technical Assistance.
10464	LCN Node <i>nn</i> : Failed attempting to create an Event: HR = <i>nnnnn</i> . This is an Internal TPS/CLM Platform error; contact Honeywell Technical Assistance.

Appendix F – UCLM: CCP AND UCIM REDESIGNED

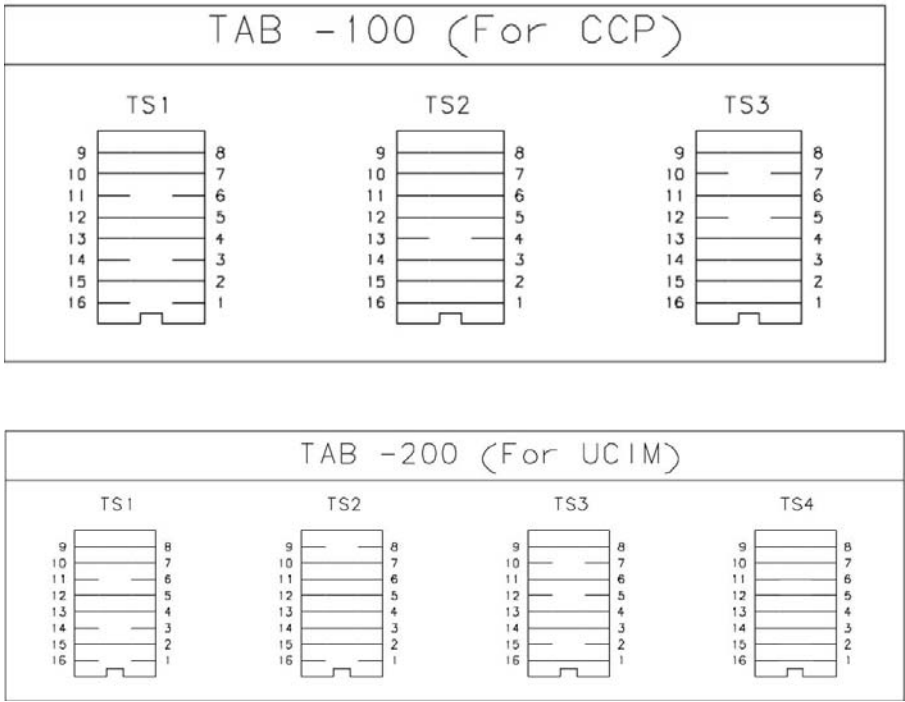
CCP (51402426) and UCIM (51402436) boards have been redesigned and are now known as UCLM-51306409. UCLM is a combination of CCP and UCIM.

Ordering Information To order the boards use the Honeywell part numbers.

Part Number	Board Name
51306409-100	CCP
51306409-200	UCIM

F.1 Switch Settings of UCLM Board

Figure F-1 Switch settings of redesigned CCP and UCIM



There is no change in the procedure to load AM personality on UCLM. Procedure remains the same as it was in CCP and UCIM.

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